

SUMMARY OF SAFETY AND EFFECTIVENESS DATA (SSED)

I. GENERAL INFORMATION

Device Generic Name:	Next generation sequencing oncology panel, somatic or germline variant detection system
Device Trade Name:	FoundationOne CDx (F1CDx)
Device Procode:	PQP
Applicant's Name and Address:	Foundation Medicine, Inc. 150 Second Street Cambridge, MA 02141
Date(s) of Panel Recommendation:	None
Premarket Approval Application (PMA) Number:	P170019/S054
Date of FDA Notice of Approval:	January 16, 2025

The original PMA (P170019) for FoundationOne CDx (F1CDx) was approved on November 30, 2017 for the detection of genetic alterations in patients who may benefit from one of eighteen FDA-approved therapies for non-small cell lung cancer (NSCLC), melanoma, breast cancer, colorectal cancer (CRC), and ovarian cancer. Subsequently, additional PMA supplements were approved for expanding the indications for use of F1CDx since the original approval. See Section VII for more details.

The current supplement was submitted to expand the indication for F1CDx to include a companion diagnostic (CDx) indication for the detection of *BRAF* V600 mutations and *BRAF* fusions in patients with pediatric low-grade glioma who may benefit from treatment with OJEMDA (tovorafenib).

II. INDICATIONS FOR USE

FoundationOne CDx (F1CDx) is a qualitative next generation sequencing based *in vitro* diagnostic test that uses targeted high throughput hybridization-based capture technology for detection of substitutions, insertion and deletion alterations (indels) and copy number alterations (CNAs) in 324 genes and select gene rearrangements, as well as genomic signatures including microsatellite instability (MSI) and tumor mutational burden (TMB) using DNA isolated from formalin-fixed paraffin embedded (FFPE) tumor tissue specimens. The test is intended as a companion diagnostic to identify

patients who may benefit from treatment with the targeted therapies listed in Table 1 in accordance with the approved therapeutic product labeling. Additionally, F1CDx is intended to provide tumor mutation profiling to be used by qualified health care professionals in accordance with professional guidelines in oncology for patients with solid malignant neoplasms. Genomic findings other than those listed in Table 1 are not prescriptive or conclusive for labeled use of any specific therapeutic product.

Table 1. Companion diagnostic indications

Tumor Type	Biomarker(s) Detected	Therapy
Breast cancer	<i>AKT1</i> E17K; <i>PIK3CA</i> R88Q, N345K, C420R, E542K, E545A, E545D, E545Q, E545K, E545G, Q546E, Q546K, Q546R, Q546P, M1043V, M1043I, H1047Y, H1047R, H1047L, and G1049R; and <i>PTEN</i> alterations	TRUQAP™ (capivasertib) in combination with FASLODEX® (fulvestrant)
	<i>ERBB2</i> (HER2) amplification	HERCEPTIN® (trastuzumab)
		KADCYLA® (ado-trastuzumab emtansine)
		PERJETA® (pertuzumab)
	<i>PIK3CA</i> C420R, E542K, E545A, E545D [1635G>T only], E545G, E545K, Q546E, Q546R, H1047L, H1047R, and H1047Y alterations	PIQRAY® (alpelisib)
Cholangiocarcinoma	<i>FGFR2</i> fusions and select rearrangements	PEMAZYRE® (pemigatinib)
Colorectal cancer	<i>KRAS</i> wild-type (absence of mutations in codons 12 and 13)	ERBITUX® (cetuximab)
	<i>KRAS</i> wild-type (absence of mutations in exons 2, 3, and 4) and <i>NRAS</i> wild-type (absence of mutations in exons 2, 3, and 4)	VECTIBIX® (panitumumab)
Melanoma	<i>BRAF</i> V600 mutation-positive	TECENTRIQ® (atezolizumab) in combination with COTELLIC® (cobimetinib) and ZELBORAF® (vemurafenib)
	<i>BRAF</i> V600E	BRAF Inhibitor Approved by FDA*
	<i>BRAF</i> V600E and V600K	BRAF/MEK Inhibitor Combinations Approved by FDA*
		MEKINIST® (trametinib)
Non-small cell lung cancer (NSCLC)	<i>ALK</i> rearrangements	ALECENSA® (alectinib)
		ALUNBRIG® (brigatinib)
		XALKORI® (crizotinib)

Tumor Type	Biomarker(s) Detected	Therapy
		ZYKADIA® (ceritinib)
	<i>BRAF</i> V600E	BRAFTOVI® (encorafenib) in combination with MEKTOVI® (binimetinib)
		TAFINLAR® (dabrafenib) in combination with MEKINIST® (trametinib)
	<i>EGFR</i> exon 19 deletions and <i>EGFR</i> exon 21 L858R alterations	EGFR Tyrosine Kinase Inhibitors (TKI) Approved by FDA*
	<i>EGFR</i> exon 20 T790M alterations	TAGRISSO® (osimertinib)
	<i>MET</i> single nucleotide variants (SNVs) and indels that lead to <i>MET</i> exon 14 skipping	TABRECTA® (capmatinib)
	<i>ROS1</i> fusions	ROZLYTREK® (entrectinib)
Ovarian cancer	<i>BRCA1</i> , <i>BRCA2</i> alterations	LYNPARZA® (olaparib)
Pediatric low-grade glioma	<i>BRAF</i> V600 mutation-positive and <i>BRAF</i> fusions	OJEMDA™ (tovorafenib)
Prostate cancer	<i>BRCA1</i> , <i>BRCA2</i> alterations	AKEEGA® (niraparib + abiraterone acetate)
		LYNPARZA® (olaparib) in combination with abiraterone
	Homologous Recombination Repair (HRR) gene (<i>BRCA1</i> , <i>BRCA2</i> , <i>ATM</i> , <i>BARD1</i> , <i>BRIP1</i> , <i>CDK12</i> , <i>CHEK1</i> , <i>CHEK2</i> , <i>FANCL</i> , <i>PALB2</i> , <i>RAD51B</i> , <i>RAD51C</i> , <i>RAD51D</i> and <i>RAD54L</i>) alterations	LYNPARZA® (olaparib)
Solid tumors	MSI-High	KEYTRUDA® (pembrolizumab)
	<i>NTRK1/2/3</i> fusions	ROZLYTREK® (entrectinib)
		VITRAKVI® (larotrectinib)
	<i>RET</i> fusions	RETEVMO® (selpercatinib)
	TMB ≥ 10 mutations per megabase	KEYTRUDA® (pembrolizumab)

*For the most current information about the therapeutic products in this group, go to:

<https://www.fda.gov/medical-devices/in-vitro-diagnostics/list-cleared-or-approved-companion-diagnostic-devices-in-vitro-and-imaging-tools>

The F1CDx assay is performed at Foundation Medicine, Inc. sites located in Cambridge, MA and Morrisville, NC.

III. CONTRAINDICATIONS

There are no known contraindications.

IV. WARNINGS AND PRECAUTIONS

The warnings/precautions and limitations are included in the FoundationOne® CDx assay labeling.

V. DEVICE DESCRIPTION

FoundationOne CDx (F1CDx) is performed at Foundation Medicine, Inc. sites located in Cambridge, MA and Morrisville, NC. The assay includes reagents, software, instruments, and procedures for testing DNA extracted from formalin-fixed, paraffin-embedded (FFPE) tumor samples.

F1CDx uses DNA extracted from formalin-fixed, paraffin-embedded (FFPE) tumor samples. The assay employs two extraction methods, (either DNAx or CoExtraction, an automated DNA/RNA co-extraction methodology) for DNA extraction from routine FFPE biopsy or surgical resection specimens; 50-1000 ng of DNA will undergo whole-genome shotgun library construction and hybridization-based capture of all coding exons from 309 cancer-related genes, one promoter region, one non-coding (ncRNA), and select intronic regions from 34 commonly rearranged genes, 21 of which also include the coding exons (refer to Table 2 and Table 3 for the complete list of genes included in F1CDx). In total, the assay detects alterations in a total of 324 genes. Using the Illumina® NovaSeq 6000, hybrid capture-selected libraries are sequenced to high uniform depth (targeting >500X median coverage with >99% of exons at coverage >100X). Sequence data is then processed using a customized analysis pipeline designed to detect all classes of genomic alterations, including base substitutions, indels, copy number alterations (amplifications and homozygous gene deletions), and select genomic rearrangements (e.g., gene fusions). Rearrangements in one of the targeted genes included in Table 3 may be reported along with their uniquely identified genomic partners, which can be any gene in the genome even if not explicitly targeted by the assay. Additionally, genomic signatures including microsatellite instability (MSI) and tumor mutational burden (TMB) are reported.

Table 2. Genes with full coding exonic regions included in F1CDx for the detection of substitutions, insertions and deletions (indels), and copy number alterations (CNAs)

<i>ABL1</i>	<i>BRAF</i>	<i>CDKN1A</i>	<i>EPHA3</i>	<i>FGFR4</i>	<i>IKZF1</i>	<i>MCL1</i>	<i>NKX2-1</i>	<i>PMS2</i>	<i>RNF43</i>	<i>TET2</i>
<i>ACVR1B</i>	<i>BRCA1</i>	<i>CDKN1B</i>	<i>EPHB1</i>	<i>FH</i>	<i>INPP4B</i>	<i>MDM2</i>	<i>NOTCH1</i>	<i>POLD1</i>	<i>ROS1</i>	<i>TGFB2</i>
<i>AKT1</i>	<i>BRCA2</i>	<i>CDKN2A</i>	<i>EPHB4</i>	<i>FLCN</i>	<i>IRF2</i>	<i>MDM4</i>	<i>NOTCH2</i>	<i>POLE</i>	<i>RPTOR</i>	<i>TIPARP</i>
<i>AKT2</i>	<i>BRD4</i>	<i>CDKN2B</i>	<i>ERBB2</i>	<i>FLT1</i>	<i>IRF4</i>	<i>MED12</i>	<i>NOTCH3</i>	<i>PPARG</i>	<i>SDHA</i>	<i>TNFAIP3</i>
<i>AKT3</i>	<i>BRIP1</i>	<i>CDKN2C</i>	<i>ERBB3</i>	<i>FLT3</i>	<i>IRS2</i>	<i>MEF2B</i>	<i>NPM1</i>	<i>PPP2R1A</i>	<i>SDHB</i>	<i>TNFRSF14</i>
<i>ALK</i>	<i>BTG1</i>	<i>CEBPA</i>	<i>ERBB4</i>	<i>FOXL2</i>	<i>JAK1</i>	<i>MEN1</i>	<i>NRAS</i>	<i>PPP2R2A</i>	<i>SDHC</i>	<i>TP53</i>
<i>ALOX12B</i>	<i>BTG2</i>	<i>CHEK1</i>	<i>ERCC4</i>	<i>FUBP1</i>	<i>JAK2</i>	<i>MERTK</i>	<i>NT5C2</i>	<i>PRDM1</i>	<i>SDHD</i>	<i>TSC1</i>

AMER1	BTK	CHEK2	ERG	GABRA6	JAK3	MET	NTRK1	PRKARIA	SETD2	TSC2
APC	C11orf30	CIC	ERF1	GATA3	JUN	MITF	NTRK2	PRKCI	SF3B1	TYRO3
AR	CALR	CREBBP	ESR1	GATA4	KDM5A	MKNK1	NTRK3	PTCH1	SGK1	U2AF1
ARAF	CARD11	CRKL	EZH2	GATA6	KDM5C	MLH1	P2RY8	PTEN	SMAD2	VEGFA
ARFRP1	CASP8	CSF1R	FAM46C	GID4 (C17orf39)	KDM6A	MPL	PALB2	PTPN11	SMAD4	VHL
ARID1A	CBFB	CSF3R	FANCA	GNA11	KDR	MRE11A	PARK2	PTPRO	SMARCA4	WHSC1
ASXL1	CBL	CTCF	FANCC	GNA13	KEAP1	MSH2	PARP1	QKI	SMARCB1	WHSC1L1
ATM	CCND1	CTNNA1	FANCG	GNAQ	KEL	MSH3	PARP2	RAC1	SMO	WT1
ATR	CCND2	CTNNB1	FANCL	GNAS	KIT	MSH6	PARP3	RAD21	SNCAIP	XPO1
ATRX	CCND3	CUL3	FAS	GRM3	KLHL6	MST1R	PAX5	RAD51	SOCS1	XRCC2
AURKA	CCNE1	CUL4A	FBXW7	GSK3B	KMT2A (MLL)	MTAP	PBRM1	RAD51B	SOX2	ZNF217
AURKB	CD22	CXCR4	FGF10	H3F3A	KMT2D (MLL2)	MTOR	PDCD1	RAD51C	SOX9	ZNF703
AXIN1	CD274	CYP17A1	FGF12	HDAC1	KRAS	MUTYH	PDCD1LG2	RAD51D	SPEN	
AXL	CD70	DAXX	FGF14	HGF	LTK	MYC	PDGFRA	RAD52	SPOP	
BAP1	CD79A	DDR1	FGF19	HNF1A	LYN	MYCL	PDGFRB	RAD54L	SRC	
BARD1	CD79B	DDR2	FGF23	HRAS	MAF	MYCN	PDK1	RAF1	STAG2	
BCL2	CDC73	DIS3	FGF3	HSD3B1	MAP2K1	MYD88	PIK3C2B	RARA	STAT3	
BCL2L1	CDH1	DNMT3A	FGF4	ID3	MAP2K2	NBN	PIK3C2G	RB1	STK11	
BCL2L2	CDK12	DOT1L	FGF6	IDH1	MAP2K4	NF1	PIK3CA	RBM10	SUFU	
BCL6	CDK4	EED	FGFR1	IDH2	MAP3K1	NF2	PIK3CB	REL	SYK	
BCOR	CDK6	EGFR	FGFR2	IGF1R	MAP3K13	NFE2L2	PIK3R1	RET	TBX3	
BCORL1	CDK8	EP300	FGFR3	IKBKE	MAPK1	NFKBIA	PIM1	RICTOR	TEK	

Table 3. Genes with select intronic regions for the detection of gene rearrangements, a promoter region, and an ncRNA gene

<i>ALK</i> introns 18, 19	<i>BRCA1</i> introns 2, 7, 8, 12, 16, 19, 20	<i>ETV4</i> introns 5, 6	<i>EZR</i> introns 9- 11	<i>KIT</i> intron 16	<i>MYC</i> intron 1	<i>NUTM1</i> intron 1	<i>RET</i> introns 7- 11	<i>SLC34A2</i> intron 4
<i>BCL2</i> 3'UTR	<i>BRCA2</i> intron 2	<i>ETV5</i> introns 6, 7	<i>FGFR1</i> intron 1, 5, 17	<i>KMT2A</i> (<i>MLL</i>) introns 6- 11	<i>NOTCH2</i> intron 26	<i>PDGFRA</i> introns 7, 9, 11	<i>ROS1</i> introns 31- 35	<i>TERC</i> ncRNA
<i>BCR</i> introns 8, 13, 14	<i>CD74</i> introns 6- 8	<i>ETV6*</i> introns 5, 6	<i>FGFR2</i> intron 1, 17	<i>MSH2</i> intron 5	<i>NTRK1</i> introns 8- 10	<i>RAF1</i> introns 4-8	<i>RSPO2</i> intron 1	<i>TERT</i> Promoter
<i>BRAF</i> introns 7- 10	<i>EGFR</i> introns 7, 15, 24-27	<i>EWSR1</i> introns 7- 13	<i>FGFR3</i> intron 17	<i>MYB</i> intron 14	<i>NTRK2</i> Intron 12	<i>RARA</i> intron 2	<i>SDC4</i> intron 2	<i>TMPRSS2</i> introns 1- 3

*ETV6 is a common rearrangement partner for *NTRK3*

Test Output

The output of the test includes:

Category 1: CDx Claims noted in Table 1 of the Intended Use

Category 2: Cancer Mutations with Evidence of Clinical Significance

Category 3: Cancer Mutations with Potential Clinical Significance

Genomic findings other than those listed in Table 1 of the intended use statement (i.e., Categories 2 and 3) are not prescriptive or conclusive for labeled use of any specific therapeutic product.

Test Kit Contents

The test includes a sample shipping kit, which is sent to ordering laboratories. The shipping kit contains the following components:

- Specimen Preparation Instructions
- Shipping Instructions
- Return Shipping Label

Instruments

The F1CDx assay is intended to be performed with serial number-controlled instruments as indicated in Table 4, below. All instruments are qualified by Foundation Medicine, Inc. (FMI) under FMI's Quality System.

Table 4. Instruments for use with the F1CDx assay

Instrument
Hamilton STAR/STARlet Liquid Handling Workstation
Covaris LE220-plus Focused ultrasonicator
Thermo Fisher Scientific KingFisher™ Flex with 96 Deep-well Head
Illumina® NovaSeq 6000 System

Test Process

All assay reagents included in the F1CDx assay process are qualified by FMI and are compliant with the medical device Quality System Regulation (QSR).

A. Specimen Collection and Preparation

Formalin-fixed, paraffin-embedded (FFPE) tumor specimens are collected and prepared following standard pathology practices. FFPE specimens may be received either as unstained slides or as an FFPE block.

Prior to starting the assay, a Hematoxylin and Eosin (H&E) stained slide is prepared, and then reviewed by a board-certified pathologist to confirm disease ontology and to ensure that adequate tissue (0.6 mm³), tumor content ($\geq 20\%$ tumor) and sufficient nucleated cells are present to proceed with the assay.

B. DNA Extraction

DNAx Extraction Method

Specimens passing pathology review are queued for DNA extraction which begins with lysis of cells from FFPE tissue by digestion with a proteinase K buffer followed by automated purification using the 96-well KingFisher™ FLEX Magnetic Particle Processor.

After completion of DNA extraction, double-stranded DNA (dsDNA) is quantified by the Quant-iT™ PicoGreen® fluorescence assay using the provided lambda DNA standards (Invitrogen) prior to Library Construction (LC). The sample must yield a minimum of 55 ng of genomic DNA to ensure sufficient DNA for quality control (QC) and to proceed with LC.

CoEx Extraction Method

Specimens passing pathology review are queued for nucleic acid extraction which begins with placement of the FFPE samples into an AutoLys tube, where using a pre-programmed automated method, the AutoLys STAR adds RNA digestion and proteinase K solutions. The RNA containing lysate is removed for downstream RNA extraction using the KingFisher RNA extraction process.

The AutoLys Tubes containing partially digested tissue then receive DNA Lysis solution on the AutoLys STAR and are placed into a Vortemp for digestion. The sample is then centrifuged to separate sample-associated paraffin from the lysate, and returned to the AutoLys STAR where the lysate is transferred to a KingFisher dKF

plate. The dKF plate is loaded onto the Hamilton STAR for automated addition of DNA binding buffer and magnetic beads, and DNA isolation is performed using the KingFisher Flex. The DNA samples are then transferred to matrix tubes on DNAE plates using Hamilton STAR before proceeding to DNA quantification.

After completion of DNA extraction, double-stranded DNA (dsDNA) is quantified by the Quant-iT™ PicoGreen® fluorescence assay using the provided lambda DNA standards (Invitrogen) prior to Library Construction (LC). The sample must yield a minimum of 55 ng of genomic DNA to ensure sufficient DNA for quality control (QC) and to proceed with LC.

C. Library Construction

Library Construction (LC) begins with the normalization of DNA to 50-1000 ng. The normalized DNA samples are randomly sheared (fragmented) to ~200 bp by adaptive focused acoustic sonication using the Covaris LE220 before purification with a 1.8X volume of AMPure® XP Beads (Agencourt®). Solid-phase reversible immobilization (SPRI) purification and subsequent library construction with the NEBNext® reagents (custom-filled kits by New England Biolabs), including mixes for end repair, dA addition and ligation, are performed in 96-well plates (Eppendorf) on the Bravo Benchbot (Agilent) or Hamilton Microlab STAR/STARlet Liquid Handling Workshop using the “with-bead” protocol to maximize reproducibility and library yield. Indexed (6 bp barcodes) sequencing libraries are polymerase chain reaction (PCR) amplified with HiFi™ (Kapa) for 10 cycles, and subsequently 1.8X SPRI purified. Purification and dilution for QC are performed.

Following LC, a QC procedure is performed by quantifying single-stranded DNA (ssDNA) from purified libraries using the Quant-iT™ PicoGreen® ssDNA Assay Kit (Life Technologies) read on a Molecular Devices Multimode SpectraMax M2 plate Reader. Libraries yielding insufficient sequencing library are failed.

D. Hybrid Capture

Hybrid Capture (HC) begins with normalization of each library to 500-2000 ng. Normalized samples then undergo solution hybridization which is performed using a > 50-fold molar excess of a pool of individually synthesized 5'-biotinylated DNA 120 bp oligonucleotides. The baits target ~1.8 Mb of the human genome including all coding exons of 309 cancer-related genes, introns or non-coding regions of 35 genes, plus > 3,500 single nucleotide polymorphisms (SNPs) located throughout the genome. Baits are designed by tiling overlapping 120 bp DNA sequence intervals covering target exons (60 bp overlap) and introns (20 bp overlap), with a minimum of three baits per target; SNP targets are allocated one bait each. Intronic baits are filtered for repetitive elements² as defined by the UCSC Genome RepeatMasker track.

After hybridization, the library-bait duplexes are captured on paramagnetic MyOne™ streptavidin beads (Invitrogen), and off-target material is removed by washing one time with 1X SSC at 25°C and four times with 0.25X SSC at 55°C. The PCR master mix is added to directly amplify (12 cycles) the captured library from the washed

beads.³ After 12 cycles of amplification, the samples are 1.8X SPRI purified. Purification and dilution for QC are performed.

QC for HC is performed by measuring dsDNA yield using the Quant-iT™ PicoGreen® dsDNA Assay Kit (Life Technologies) read on a Molecular Devices Multimode SpectraMax M2 plate Reader. Captured libraries yielding less than 140 ng of sequencing library are failed.

E. Sequencing

Sequencing is performed using on-board cluster generation using patterned flow cell technology to generate monoclonal clusters from a single DNA template followed by sequencing using sequencing by synthesis (SBS) chemistry on the Illumina NovaSeq 6000. Fluorescently labeled 3'-blocked deoxynucleotide triphosphates (dNTPs) together with a polymerase are incorporated through the flow cell to create a growing nucleotide chain that is excited by a laser. A camera captures the emission color of the incorporated base and then is cleaved off. The terminator is then removed to allow the nucleotide to revert to its natural form and to allow the polymerase to add another base to the growing chain. A new pool of fluorescently labeled 3'-blocked dNTPs are added with each new sequencing cycle. The color changes for each new cycle as a new base is added to the growing chain. This method allows for millions of discrete clusters of clonal copies of DNA to be sequenced in parallel.

F. Sequence Analysis

Sequence data is analyzed using proprietary software developed by FMI. Sequence data are mapped to the human genome (hg19) using Burrows-Wheeler Aligner (BWA) v0.5.9.⁴ PCR duplicate read removal and sequence metric collection are performed using Picard 1.47 (<http://picard.sourceforge.net>) and SAMtools 0.1.12a.⁵ Local alignment optimization is performed using Genome Analysis Toolkit (GATK) 1.0.4705.⁶ Variant calling is performed only in genomic regions targeted by the test.

Base substitution detection is performed using a Bayesian methodology, which allows for the detection of novel somatic alterations at low mutant allele frequency (MAF) and increased sensitivity for alterations at hotspot sites through the incorporation of tissue-specific prior expectations.⁷ Reads with low mapping (mapping quality < 25) or base calling quality (base calls with quality ≤ 2) are discarded. Final calls are made at MAF ≥ 5% (MAF ≥ 1% at hotspots).

To detect indels, *de novo* local assembly in each targeted exon is performed using the de-Bruijn approach.⁸ Key steps are:

- Collecting all read pairs for which at least one read maps to the target region.
- Decomposing each read into constituent k-mers and constructing an enumerable graph representation (de-Bruijn) of all candidate non-reference haplotypes present.

- Evaluating the support of each alternate haplotype with respect to the raw read data to generate mutational candidates. All reads are compared to each of the candidate haplotypes via ungapped alignment, and a read ‘vote’ for each read is assigned to the candidate with best match. Ties between candidates are resolved by splitting the read vote, weighted by the number of reads already supporting each haplotype. This process is iterated until a ‘winning’ haplotype is selected.
- Aligning candidates against the reference genome to report alteration calls.

Filtering of indel candidates is carried out similarly to base substitutions, with an empirically increased allele frequency threshold at repeats and adjacent sequence quality metrics as implemented in GATK: % of neighboring bases mismatches < 25%, average neighboring base quality > 25, average number of supporting read mismatches ≤ 2 . Final calls are made at MAF $\geq 5\%$ (MAF $\geq 3\%$ at hotspots).

Copy number alterations (CNAs) are detected using a comparative genomic hybridization (CGH)-like method. First, a log-ratio profile of the sample is acquired by normalizing the sequence coverage obtained at all exons and genome-wide SNPs (~3,500) against a process-matched normal control. This profile is segmented and interpreted using allele frequencies of sequenced SNPs to estimate tumor purity and copy number at each segment. Amplifications are called at segments with ≥ 6 copies or ≥ 7 for triploid/ ≥ 8 for tetraploid tumors) and homozygous deletions at 0 copies, in samples with tumor purity $\geq 20\%$. Amplifications in *ERBB2* are called positive at segments with ≥ 5 copies for diploid tumors.

Genomic rearrangements are identified by analyzing chimeric read pairs. Chimeric read pairs are defined as read pairs for which reads map to separate chromosomes, or at a distance of over 10 megabase (Mb). Pairs are clustered by genomic coordinate of the pairs, and clusters containing at least five chimeric pairs (three for known fusions) are identified as rearrangement candidates. Filtering of candidates is performed by mapping quality (average read mapping quality in the cluster must be 30 or above) and distribution of alignment positions. Rearrangements are annotated for predicted function (e.g., creation of fusion gene).

To determine microsatellite instability (MSI) status, F1CDx employs a fraction based (FB) MSI algorithm to categorize a tumor specimen as MSI-High (MSI-H) or microsatellite stable (MSS). The FB-MSI algorithm calculates the fraction of microsatellite loci determined to be altered or unstable (i.e., the fraction unstable loci score) based on a genome-wide analysis across >2000 microsatellite loci. For a given microsatellite locus, non-somatic alleles are discarded, and the microsatellite is categorized as unstable if remaining alleles differ from the reference genome. The final fraction unstable loci score is calculated as the number of unstable microsatellite loci divided by the number of evaluable microsatellite loci. Two FB-MSI score thresholds are applied to classify a tumor specimen as having MSI-H or MSS status. MSI-H status is reported for patients with solid tumors whose samples have FB-MSI scores ≥ 0.0124 while MSS status is reported for patients with solid tumors whose samples have FB-MSI scores ≤ 0.0041 . Per the F1CDx assay, a patient whose tumor

has an MSI-H score ≥ 0.0124 is reported as eligible for treatment with KEYTRUDA. For patients with solid tumors whose samples have FB-MSI scores >0.0041 and <0.0124 , an MSI “Cannot be Determined” result is reported. Patients with this result should be re-tested with a validated orthogonal (alternative) method as these MSI scores represent a range of scores with low reliability. Patients with solid tumors may also receive an MSI status reported as MSI-Cannot Be Determined due to a quality control (QC) failure. Patients with this result should consider re-testing with FoundationOne CDx or an orthogonal (alternative) method, if clinically appropriate.

Tumor mutational burden (TMB) is measured by counting all synonymous and non-synonymous substitution and indel variants present at 5% allele frequency or greater and filtering out potential germline variants according to published databases of known germline polymorphisms including Single Nucleotide Polymorphism database (dbSNP) and Exome Aggregation Consortium (ExAC). Additional germline alterations still present after database querying are assessed for potential germline status and filtered out using a somatic-germline/zygosity (SGZ) algorithm. Furthermore, known and likely driver mutations are filtered out to exclude bias of the data set. The resulting mutation number is then divided by the coding region corresponding to the number of total variants counted, or 793 kb. The resulting number is communicated as mutations per Mb unit (mut/Mb).

After completion of the Analysis Pipeline, variant data are displayed in the FMI custom developed CATi software applications with sequence QC metrics. As part of data analysis QC for every sample, the F1CDx assay assesses cross-contamination through the use of a SNP profile algorithm, reducing the risk of false-positive calls that could occur as a result of an unexpected contamination event. Sequence data are reviewed by trained bioinformatics personnel. Samples failing any QC metrics are automatically held and not released.

G. Report Generation

Approved results are annotated by automated software with CDx relevant information and are merged with patient demographic information and any additional information provided by FMI as a professional service prior to approval and release by the laboratory director or designee.

H. Internal Process Controls Related to the System

Positive Control

Each assay run includes a control sample run in duplicate. The control sample contains a pool of ten HapMap cell lines and is used as a positive mutation detection control. One hundred (100) different germline SNPs present across the entire targeted region are required to be detected by the analysis pipeline. If SNPs are not detected as expected, this results in a QC failure, as it indicates a potential processing error.

Sensitivity Control

The HapMap control pool used as the positive control is prepared to contain variants at 5%-10% MAF which must be detected by the analysis pipeline to ensure the expected sensitivity for each run.

Negative Control

Samples are barcoded molecularly at the LC stage. Only reads with a perfect molecular barcode sequence are incorporated into the analysis. The Analysis Pipeline includes an algorithm that analyzes the SNP profile of each specimen to identify potential contamination that may have occurred prior to molecular barcoding and can detect contamination lower than 1%.

I. Variant Classification

Biomarker Rules for SNVs and indels that lead to *MET* exon 14 skipping

An SNV or indel in *MET* shall be considered to result in skipping of exon 14 if one or more of the following criteria are met:

1. Deletions greater than or equal to 5 bp that affect positions -3 to -30 in the intronic region immediately adjacent to the splice acceptor site at the 5'boundary of *MET* exon 14.
2. Indels affecting positions -1 or -2 at the splice acceptor site of the 5'boundary of *MET* exon 14.
3. Base substitutions and indels affecting positions 0, +1, +2, or +3 at the splice donor site of the 3'boundary of *MET* exon 14.

Homologous Recombination Repair (HRR) Genes

A clinical report is provided to the ordering physician for each F1CDx test performed at Foundation Medicine, Inc. Each report is generated and reviewed by an internal team consisting of clinical bioinformatics analysts, scientists, curators, and pathologists for mutations positive for the therapies identified. Each sample is assessed for mutations in the 14 HRR genes, *ATM*, *BARD1*, *BRCA1*, *BRCA2*, *BRIP1*, *CDK12*, *CHEK1*, *CHEK2*, *FANCL*, *PALB2*, *RAD51B*, *RAD51C*, *RAD51D*, and *RAD54L* (Table 5). For these genes, both deleterious and suspected deleterious mutations in short variant, copy number alteration, and rearrangement variant classes are determined by an in-house software pipeline. Alterations listed in the COSMIC database and homozygous deletions are considered deleterious. Suspected deleterious mutations include truncating events (i.e., splice, frameshift, and nonsense alterations), as well as large rearrangements that disrupt the coding sequence. The COSMIC check is a second layer of check for HRR positive suspected deleterious alterations. All splice, nonsense, and frameshift alterations in HRR genes are considered biomarker positive and would be considered as suspected deleterious mutations (or “likely” status in FMI reporting rules). If these mutations are additionally reported in COSMIC, they would be listed as deleterious mutations (or “known” status in FMI reporting).

The F1CDx assay is intended as an aid in selecting prostate cancer patients with deleterious or suspected deleterious HRR variants, identified by the rules below, and who may be eligible for treatment with Lynparza® (olaparib).

Table 5. Mutation types identified in the HRR genes

Variant Class	Alteration Type	Description*
Short Variant	Nonsense, frameshift or splice site	Any deleterious nonsense, frameshift, or splicing event that spans or occurs within ± 2 bases of the intron/exon junction
	Missense or non-frameshift	Any of the mutations listed in Table 6 for <i>ATM</i> , <i>BRCA1</i> , and <i>BRCA2</i>
Copy Number Alteration	Homozygous copy number loss	Deleterious homozygous copy number loss of one or more exons
Rearrangement	Rearrangement	Any rearrangement that disrupts protein function

*For *BRCA2*, truncating mutations must occur upstream of bases encoding amino acid 3326. Additionally, the frameshift mutation T367fs*13 in *FANCL* is ineligible. All short variants must occur in the canonical transcript.

The specific deleterious mutation (DM) and suspected deleterious mutation (SDM) missense mutations or non-frameshift mutations for *BRCA1*, *BRCA2*, and *ATM* are shown in Table 6-8, below. However, any missense or non-frameshift mutations in the other 12 genes would not be considered HRR positive.

Table 6. Eligible deleterious mutations in the *ATM* gene (for olaparib CDx claim only)

M1I	Y2470D	2832C
M1L	R2547_S2549del	S2855_V2856>RI
M1T	A2622V	D2913Y
P292L	D2625_A2626>EP	R3008C
D2016G	D2708N	R3008H
R2032K	V2716A	splice site 331 +5G>A
A2067D	G2765S	splice site 8418+5_8418+8delGTGA
R2227C	F2827C	

Table 7. List of short variants in *BRCA1*

M1R	C44S	R1495T	D1692Y*	G1738E	Y1853C
M1I	C44Y	E1559K	C1697R	G1738R	C1787_G1788>SD
M1V	C47F	E1559Q	R1699Q	L1764P	Splice site 212+3A>G
M1T	C61G	A1623G	R1699W	I1766S	Splice site 213-11T>G
M18T	C61Y*	S1655F	L1705P	G1770V	Splice site 213-12A>G
L22S	C64G	T1685A	G1706E	M1775K	Splice site 302-2C>G
C24R	C64R	T1685I	G1706R	M1775R	Splice site 4986+3G>C
T37K	C64W*	H1686R	A1708E	L1780P	Splice site 4986+5G>A
C39G*	C64Y	V1688del	S1715N	G1788V	Splice site 4986+6T>G
C39R	R71K	M1689R	S1715R	P1812A	Splice site 4986+6T>C
C39W*	R71G	T1691I	W1718C	V1833M	Splice site 5074+3A>G
C39Y	R170Q*	T1691K	S1722F	W1837R	Splice site 5194-12G>A
H41R	R1495K	D1692N	V1736A	W1837C	Splice site 5406+4A>G
C44F	R1495M	D1692H	V1736G	V1838E	

*variants are part of the biomarker definition for the niraparib CDx claim only

Table 8. List of short variants in *BRCA2*

M1R	R2336L*	R2659G	L2686P	Y2726C	N3124I
M1I	R2336P	R2659K	L2686P	G2748D	Splice site 316+4delA
S142I*	L2510P	R2659T	T2722R	G2793R	Splice site 316+5G>A
V159M	H2623R	Y2660D	D2723A	E3002K	Splice site 8487+3A>G
V211I	W2626C	E2663V	D2723G	R3052W	Splice site 8754+4A>G
V211L	I2627F	S2670L	D2723H	G3076V	Splice site 8754+5G>A
R2336H	L2653P	I2675V	D2723V	D3095E	Splice site 8754+3G>C

*variants are part of the biomarker definition for the niraparib CDx claim only

Biomarker Rules for Rearrangements that Lead to *NTRK1*, *NTRK2*, or *NTRK3* Fusions:

Rearrangements in *NTRK1*, *NTRK2*, or *NTRK3* shall be considered CDx biomarker positive, that is, to lead to a *NTRK1*, *NTRK2*, or *NTRK3* RNA fusion, if the following criterion is met:

- In-strand rearrangement events that may lead to an *NTRK1*, *NTRK2* or *NTRK3* RNA fusion with a previously reported or novel partner gene in which the kinase domain is not disrupted. This also includes rearrangement events that result in reciprocal fusions (*NTRK*-3' and 5'-*NTRK* events).

In this regard out-of-strand events are considered as non-fusion rearrangements and are classified as CDx biomarker negative. Intragenic fusions in which genomic rearrangement events are wholly internal to the *NTRK1*, *NTRK2*, or *NTRK3* genes (i.e., *NTRK1*-*NTRK1*, *NTRK2*-*NTRK2*, *NTRK3*-*NTRK3* events) are also considered biomarker negative.

Unidentified partners (encoded as N/A) or LINC non-coding partners are also considered CDx biomarker negative.

Biomarker Rules for ALK Rearrangements:

Rearrangements in *ALK* shall be considered CDx biomarker positive if the following criterion is met:

- Any oncogenic *ALK* rearrangement whose breakpoint occurs within ALK intron 19 or whose partner gene is EML4.

Biomarker Rules for FGFR2 Fusions and Select Rearrangements:

Rearrangements in *FGFR2* shall be considered CDx biomarker positive if the following criteria are met:

- The rearrangement event involved *FGFR2* and a literature-derived known partner gene regardless of strand or frame,
- The rearrangement event involved *FGFR2* and a novel partner gene that is both in-frame and in-strand,
- Any *FGFR2* rearrangement with one breakpoint in the hotspot region (intron 17-18) and the other breakpoint in the intergenic region or within another gene. This rule excludes 3' duplications of only exon 18,
- Intragenic duplication of kinase domain (exon 9-17).

Biomarker Rules for Rearrangements that Lead to ROS1 Fusions:

Rearrangements in *ROS1* shall be considered CDx biomarker positive, i.e., to lead to *ROS1* RNA fusion, if the following condition is met:

- In-strand rearrangement events that may lead to a *ROS1* RNA fusion with another protein coding gene in which the *ROS1* kinase domain is not disrupted. *ROS1* must be on the 3' end of the detected fusion.

In this regard, out-of-strand events are considered as non-fusion rearrangements and are classified as CDx biomarker negative. Intragenic fusions in which genomic rearrangement events are wholly internal to the *ROS1* (i.e., *ROS1-ROS1* events) are also considered biomarker negative. Unidentified partners (encoded as N/A) or LINC non-coding partners are also considered CDx biomarker negative. *ROS1* fusions with novel partners are required to be in frame.

Biomarker Rules for RET Fusions:

Fusions in *RET* shall be considered CDx biomarker positive if the following criteria are met:

- Any fusion event involving *RET* and another protein-coding gene
- *RET* and the partner gene must be in the same 5'-3' orientation
- *RET* must be on the 3' end of the detected rearrangement
- The *RET* breakpoint must occur before the start of the kinase domain (amino acids 724-1016)

Biomarker Rule for AKT1, PIK3CA, and PTEN Alterations:

Alterations in *AKT1*, *PIK3CA*, and/or *PTEN* are considered Companion biomarker positive if the following criteria are met.

Table 9 Biomarker definition for *AKT1*, *PIK3CA*, and *PTEN*

Gene (Transcript)	Alteration Type	Protein/Transcript Effect
<i>AKT1</i> (NM_001014431)	SNV	E17K
<i>PIK3CA</i> (NM_006218)	SNV	R88Q, N345K, C420R, E542K, E545A, E545D, E545Q, E545K, E545G, Q546E, Q546K, Q546R, Q546P, M1043V, M1043I, H1047Y, H1047R, H1047L, and G1049R
<i>PTEN</i> (NM_000314)	SNV and indels	C124R, G129E, R130Q, C136R, S170R, R173C, C124S, G129V, R130G, C136Y, G129R, R130L, R130P Any nonsense, frameshift, or splice site alteration Missense mutations in the start codon and short variant deletions spanning from upstream of the start codon (annotated as M1?) are included in this category
	Copy Number Alteration	Homozygous deletion (HD) represents a deletion of one or more exons regardless of transcript in both alleles
	Rearrangement (RE)	Any rearrangement that disrupts protein function, regardless of transcript - Intragenic events including duplications of only part of the gene, deletions, or inversions. - Translocations, deletions, or inversions where one breakpoint is in <i>PTEN</i> and the other breakpoint is in another gene or intergenic region.

Biomarker Rule for *BRAF* V600 mutations and *BRAF* fusions:

BRAF V600 and *BRAF* fusion alterations are considered companion diagnostic biomarker positive if the following criteria are met.

Table 10. Biomarker definition for *BRAF* V600X and *BRAF* fusion alterations

Gene (Transcript)	Variant Type	Biomarker rules
<i>BRAF</i> (NM_004333)	Short Variant	Any missense alteration resulting in V600X ¹
	Rearrangement	Any fusion event that involves <i>BRAF</i> and another protein-coding gene and meets all the following criteria: • <i>BRAF</i> and the partner gene must be in the same 5'-3' orientation • <i>BRAF</i> must be on the 3' end of the detected fusion • The <i>BRAF</i> breakpoint must occur after the autoinhibitory domain and before the kinase domain

¹X refers to any single amino acid change resulting from a missense alteration.

VI. ALTERNATIVE PRACTICES AND PROCEDURES

There are FDA-approved companion diagnostic (CDx) alternatives for the detection of genetic alterations using FFPE tumor specimens, as listed in Table 1 of the F1CDx intended use statement. The approved CDx tests are listed in Table 11 below; for additional details see FDA List of Cleared or Approved Companion Diagnostic Devices at: <https://www.fda.gov/medical-devices/vitro-diagnostics/list-cleared-or-approved-companion-diagnostic-devices-vitro-and-imaging-tools>. Each alternative has its own advantages and disadvantages. A patient should fully discuss these alternatives with his/her physician to select the method that best meets expectations and lifestyle.

Table 11. List of FDA approved CDx assays for genes targeted by F1CDx

	Device	Company	Technology	Therapy	Indication
HER2-Amplification	PathVysion HER-2 DNA Probe Kit	Abbott Molecular, Inc.	FISH	HERCEPTIN (trastuzumab)	Breast cancer
	PATHWAY Anti-HER-2/neu (4B5) Rabbit Monoclonal Primary Antibody	Ventana Medical Systems, Inc.	IHC	HERCEPTIN (trastuzumab)	Breast cancer
	InSite HER-2/neu Kit	Biogenex Laboratories, Inc.	IHC	HERCEPTIN (trastuzumab)	Breast cancer
	SPOT-Light HER2 CISH Kit	Life Technologies, Inc.	CISH	HERCEPTIN (trastuzumab)	Breast cancer
	Bond Oracle HER2 IHC System	Leica Biosystems	IHC	HERCEPTIN (trastuzumab)	Breast cancer
	HER2 CISH pharmDx Kit	Dako Denmark A/S	CISH	HERCEPTIN (trastuzumab)	Breast cancer
	INFORM HER2 Dual ISH DNA Probe Cocktail	Ventana Medical Systems, Inc.	Dual ISH	HERCEPTIN (trastuzumab)	Breast cancer

	Device	Company	Technology	Therapy	Indication
	HercepTest	Dako Denmark A/S	IHC	HERCEPTIN (trastuzumab) PERJETA (pertuzumab) KADCYLA (ado-trastuzumab emtansine)	Breast cancer Gastric or Gastroesophageal junction adenocarcinoma
	HER2 FISH pharmDx Kit	Dako Denmark A/S	FISH	HERCEPTIN (trastuzumab) PERJETA (pertuzumab) KADCYLA (ado-trastuzumab emtansine)	Breast cancer Gastric or Gastroesophageal junction adenocarcinoma
BRAF-V600E and V600K	THxID BRAF Kit	bioMerieux	PCR	MEKINIST (trametinib)	Melanoma
	cobas 4800 BRAF V600 Mutation Test	Roche Molecular Systems, Inc.	PCR	COTELLIC (cobimetinib) ZELBORAF (vemurafenib)	Melanoma
BRAF-V600E	cobas 4800 BRAF V600 Mutation Test	Roche Molecular Systems, Inc.	PCR	ZELBORAF (vemurafenib)	Melanoma
	THxID BRAF Kit	bioMerieux	PCR	TAFINLAR (dabrafenib)	Melanoma
	Oncomine Dx Target Test	Life Technologies, Inc.	NGS	TAFINLAR (dabrafenib) MEKINIST (trametinib)	NSCLC
	<i>therascreen</i> BRAF V600E RGQ PCR Kit	QIAGEN	PCR	BRAFTOVI (encorafenib) Erbitux (cetuximab)	CRC
NRAS	Praxis Extended RAS Panel	Illumina, Inc.	NGS	VECTIBIX (panitumumab)	CRC
KRAS	cobas KRAS Mutation Test	Roche Molecular Systems, Inc.	PCR	ERBITUX (cetuximab) VECTIBIX (panitumumab)	CRC
	<i>therascreen</i> KRAS RGQ PCR Kit	QIAGEN	PCR	ERBITUX (cetuximab) VECTIBIX (panitumumab)	CRC
	Praxis Extended RAS Panel	Illumina, Inc.	NGS	VECTIBIX (panitumumab)	CRC

	Device	Company	Technology	Therapy	Indication
ALK – fusion	Vysis ALK Break Apart FISH Probe Kit	Abbott Molecular, Inc.	FISH	XALKORI (crizotinib)	NSCLC
	ALK (D5F3) CDx Assay	Ventana Medical Systems, Inc.	IHC	XALKORI (crizotinib)	NSCLC
EGFR – Exon 19 deletions & L858R	cobas EGFR Mutation Test v2	Roche Molecular Systems, Inc.	PCR	TARCEVA (erlotinib) TAGRISSO (osimertinib) IRESSA (gefitinib)	NSCLC
	<i>therascreen</i> EGFR RGQ PCR Kit	QIAGEN	PCR	GILOTRIF (afatinib) IRESSA (gefitinib)	NSCLC
	Oncomine Dx Target Test	Life Technologies, Inc.	NGS	IRESSA (gefitinib)	NSCLC
EGFR T790M	cobas EGFR Mutation Test v2	Roche Molecular Systems, Inc.	PCR	TAGRISSO (osimertinib)	NSCLC
BRCAl/2	BRACAnalysis CDx	Myriad Genetic Laboratories, Inc.	NGS	LYNPARZA (olaparib) LYNPARZA (olaparib)-treatment/maintenance TALZENNA (talazoparib)	Breast, pancreatic, and prostate cancers Ovarian cancer Breast cancer
	Myriad myChoice® CDx	Myriad Genetic Laboratories, Inc.	NGS	ZEJULA (niraparib) or Lynparza (olaparib)	Ovarian cancer
PIK3CA	<i>therascreen</i> PIK3CA RGQ PCR Kit	QIAGEN	PCR	PIQRAY (alpelisib)	Breast cancer
ROS1	Oncomine Dx Target Test	Life Technologies, Inc.	NGS	XALKORI (crizotinib)	NSCLC
RET	Oncomine Dx Target Test	Life Technologies, Inc.	NGS	RETEVMO (selpercatinib)	NSCLC and Thyroid Cancer
				GAVRETO (pralsetinib)	NSCLC

Abbreviations: FISH – fluorescence in situ hybridization; IHC – immunohistochemistry; CISH chromogenic in situ hybridization; ISH – in situ hybridization; PCR – polymerase chain reaction; NGS – next generation sequencing.

VII. MARKETING HISTORY

Foundation Medicine, Inc. indicates the F1CDx assay was offered as a laboratory developed test (LDT) with the name “F1 LDT” since 2012. As described in the preamble to FDA’s LDT final rule, FDA uses the term "IVDs offered as LDTs" for IVDs that are manufactured and offered as LDTs by laboratories that are certified under CLIA and that meet the regulatory requirements under CLIA to perform high complexity testing, and used within such laboratories, even if those IVDs do not fall within FDA’s traditional understanding of an LDT because they are not designed, manufactured, and used within a single laboratory.

The F1CDx Premarket Approval (PMA) was originally approved on November 30, 2017 by FDA (P170019) and is commercially available in the U.S. since March 30, 2018. The approved PMA supplements that affected the Intended Use are listed in Table 12.

Table 12: Marketing History

Submission No.	Date of Approval	Biomarker/Update	Patient Population	Drug
P170019/S004	July 1, 2019	<i>BRCA1/2</i> alterations	Ovarian Cancer	LYNPARZA® (olaparib)
P170019/S005	April 10, 2019	genomic loss of heterozygosity (LOH)	Ovarian Cancer	N/A
P170019/S006	December 3, 2019	<i>PIK3CA</i> alterations	Breast Cancer	PIQRAY® (alpelisib)
P170019/S008	July 1, 2019	<i>EGFR</i> exon 19 deletions and <i>EGFR</i> exon 21 L858R alterations	Non-Small Cell Lung Cancer	TAGRISSO® (osimertinib)
P170019/S011	May 6, 2020	<i>MET</i> single nucleotide variants (SNVs) and indels that lead to <i>MET</i> exon 14 skipping	Non-Small Cell Lung Cancer	TABRECTA® (capmatinib)
P170019/S013	April 17, 2020	<i>FGFR2</i> fusions	Cholangiocarcinoma	PEMZIRE® (pemigatinib)
P170019/S015	May 19, 2020	mutations in homologous recombination repair (HRR) genes	metastatic castration resistant prostate cancer (mCRPC)	LYNPARZA® (olaparib)

Submission No.	Date of Approval	Biomarker/Update	Patient Population	Drug
P170019/S016	June 16, 2020	high tumor mutational burden (TMB) at the cut-off of 10 mutations per megabase (mut/Mb)	Solid Tumors	KEYTRUDA® (pembrolizumab)
P170019/S017	October 23, 2020	<i>NTRK1</i> , <i>NTRK2</i> , or <i>NTRK3</i> fusions	Solid Tumors	VITRAKVI® (larotrectinib)
P170019/S021	May 28, 2021	<i>FGFR2</i> Fusion/Rearrangements	Cholangiocarcinoma	TRUSELTIQ® (infigratinib)
P170019/S022	July 21, 2021	Additional variants to <i>BRCA1</i> and <i>BRCA2</i>	Ovarian Cancer	LYNPARZA® (olaparib) or RUBRACA® (rucaparib)
		Additional variants to <i>BRCA1</i> , <i>BRCA2</i> and <i>ATM</i>	Prostate Cancer	LYNPARZA® (olaparib)
P170019/S023	June 30, 2021	<i>ALK</i> Rearrangements	Non-Small Cell Lung Cancer	ALUNBRIG® (brigatinib)
P170019/S025	November 10, 2021	<i>BRAF</i> V600E Alterations	Melanoma	BRAF Inhibitor Monotherapy Group Claim
		<i>BRAF</i> V600E or V600K Alterations	Melanoma	BRAF/MEK Inhibitor Combination Group Claim
P170019/S029	February 18, 2022	Microsatellite Instability High (MSI-H) Status	Solid Tumors	KEYTRUDA® (Pembrolizumab)
P170019/S030	January 19, 2022	<i>BRAF</i> V600 Mutation-Positive	Unresectable Or Metastatic Melanoma	TECENTRIQ® (atezolizumab) In Combination with Cobimetinib and Vemurafenib
P170019/S033	March 16, 2022	<i>EGFR</i> Exon 19 Deletions or <i>EGFR</i> Exon 21 L858R Mutations	Non-Small Cell Lung Cancer	Any One of The FDA-Approved <i>EGFR</i> Tyrosine Kinase Inhibitors (TKI)
P170019/S014	June 7, 2022	<i>NTRK1</i> , <i>NTRK2</i> , or <i>NTRK3</i> fusions	Solid Tumors	ROZLYTREK® (entrectinib)
		<i>ROS1</i> fusions	NSCLC	
P170019/S042	August 11, 2023	<i>BRCA1</i> , <i>BRCA2</i> alterations	Prostate Cancer	AKEEGA® (niraparib + abiraterone acetate)
P170019/S043	October 6, 2023	<i>RET</i> fusions	Solid Tumors	RETEVMO® (selpercatinib)
P170019/S039	October 11, 2023	<i>BRAF</i> V600E	NSCLC	BRAFTOVI® (encorafenib) in combination with MEKTOVI® (binimetinib)
P170019/S048	November 16, 2023	<i>AKT1</i> , <i>PIK3CA</i> , <i>PTEN</i> alterations	Breast Cancer	TRUQAP™ (capivasertib) in combination with

Submission No.	Date of Approval	Biomarker/Update	Patient Population	Drug
				FASLODEX® (fulvestrant)
P170019/S052	August 30, 2024	BRCA1, BRCA2 alterations	Prostate Cancer	LYNPARZA® (olaparib) in combination with abiraterone

VIII. POTENTIAL ADVERSE EFFECTS OF THE DEVICE ON HEALTH

Failure of the device to perform as expected or failure to correctly interpret test results may lead to incorrect test results, and subsequently, inappropriate patient management decisions. Patients with false positive results may undergo treatment with one of the therapies listed in the above intended use statement without clinical benefit and may experience adverse reactions associated with the therapy. Patients with false negative results may not be considered for treatment with the indicated therapy. There is also a risk of delayed results, which may lead to delay of treatment with the indicated therapy. For the specific adverse events related to the approved therapeutics, please see the approved drug product labels.

IX. SUMMARY OF NON-CLINICAL STUDIES

A. Laboratory Studies

The primary evidence for supporting the performance of F1CDx in the detection of *BRAF* V600 mutations and *BRAF* fusions in pediatric low-grade glioma (pLGG) patients was from data generated using glioma specimens within the validation studies described herein. Given the low prevalence of pLGG, for analytical studies, while prioritizing the inclusion of low-grade gliomas and ≤ 25 -year-old patient samples, *BRAF* mutation and fusion samples from both low- and high-grade gliomas and from all patient ages were used. In addition to the existing platform-level validation results (P170019), refer to Section IX.A. in P170019 Summary of Safety and Effectiveness Data, wherein, analytical concordance, site-to-site precision, limit of blank (LoB), and limit of detection (LoD) studies were conducted to support the indication for *BRAF* V600 mutations and *BRAF* fusions.

Glioma samples evaluated in the analytical validation studies (site-to-site precision, LoB, and LoD) were extracted with DNAX extraction method using the HiSeq sequencer configuration. The F1CDx device was updated to include addition of an automated DNA/RNA CoExtraction methodology (CoEx method) to enable isolation of DNA and RNA from the same FFPE tumor specimens. The F1CDx was authorized for DNA only when using the CoEx method. The F1CDx device was also updated to replace the Illumina HiSeq 4000 sequencer with the Illumina NovaSeq 6000 sequencer. Analytical validation studies were conducted to demonstrate the comparable performance between the two nucleic acid extraction methods and two sequencer configurations, using samples from multiple disease ontologies to provide evidence of

the robust detection of *BRAF* fusions and *BRAF* V600 mutations utilizing the NovaSeq configuration and the CoEx extraction method. The reproducibility and repeatability for *BRAF* mutations from the CoEx precision study included 5 *BRAF* V600 mutation samples. The aggregated reproducibility for *BRAF* mutations was 100.00% (104/104, 95% CI: [96.44%, 100.00%]) and the aggregated repeatability for *BRAF* mutations was 100.00% (50/50, 95% CI: [92.87%, 100.00%]). The CoExtraction concordance study included 19 samples with *BRAF* mutations, and 1 sample with *BRAF* fusion. The aggregated positive percent agreement (PPA) for *BRAF* mutation was 100.00% (19/19, 95% CI: [83.18%, 100.00%]) and the aggregated negative percent agreement (NPA) for *BRAF* mutation was 100.00% (20/20, 95% CI: [83.89%, 100.00%]). The aggregated PPA for *BRAF* mutation from the NovaSeq migration study using 19 samples with *BRAF* mutations, and 1 sample with *BRAF* fusion was 100.00% (19/19, 95% CI: [83.18%, 100.00%]) and the aggregated NPA for *BRAF* mutation was 100.00% (108/108, 95% CI: [96.57%, 100.00%]).

The F1CDx analytical pipeline was modified after the analytical and clinical validation studies were completed. The analytical and clinical validation provided to support the detection of *BRAF* V600 mutations and *BRAF* fusions in low-grade glioma by F1CDx were performed using the analytical pipeline versions that have undergone further iteration with modification in the final device design. To demonstrate the F1CDx analytical pipeline changes continue to support the performance for which the test is approved, regression testing using the most current analytical pipeline version will be performed post-market to confirm the robust performance for F1CDx for the detection of *BRAF* V600 mutations and *BRAF* fusions (see section XIII).

i Analytical Accuracy / Concordance

Comparison to an Orthogonal Method

To demonstrate analytical concordance, an analysis between F1CDx and an externally validated NGS assay (evNGS) to detect *BRAF* V600 mutations and *BRAF* fusions was performed. The concordance study was conducted using available residual DNA previously extracted by CoEx method from 15 FFPE tissue samples from patients with low-grade glioma (LGG) enrolled in the FIREFLY-1 (NCT04775485) clinical study that supported OJEMDA (tovorafenib) approval (Refer to Section X Summary of Primary Clinical Study for study details). The study was supplemented with DNA previously extracted using DNAX method from 189 FFPE glioma tissue samples from the FMI sample archives. In total, the available samples for the analytical concordance study included 103 *BRAF* V600 mutation and *BRAF* fusion (biomarker positive) and 101 non-altered (biomarker negative) samples.

In total, 204 samples were processed in the concordance study with representation of both *BRAF* V600 mutations and *BRAF* fusions included in the tovorafenib CDx biomarker definition. Of the 204 samples, one sample was excluded due to improper storage conditions. In addition, 201 of the remaining 203 samples were processed successfully from library construction through

sequencing by F1CDx. All 204 samples completed lab processing and passed the sample QC metrics for the evNGS assay.

A total of 203 samples were included in the concordance analysis. The F1CDx and evNGS results for the detection of *BRAF* V600 mutations and *BRAF* fusions using evNGS as the comparator assay, are provided in the contingency table in Table 13. The statistical analysis using the evNGS assay results as the reference showed a positive predictive value (PPV) of 96.88% with a 95% confidence interval (CI) of (91.21%, 98.93%) and a negative predictive value (NPV) of 97.14% with 95% CI (91.93%, 99.02%) unadjusted for prevalence, and a positive percent agreement (PPA) of 96.88% (93/96) with 95% CI (91.21%, 98.93%), and a negative percent agreement (NPA) of 97.14% with 95% CI[(91.93%, 99.02%). Since the samples were selected from different sources based on different assays, the PPA/NPA and PPV/NPV may be subject to potential bias.

Table 13. Contingency Table Comparing the Detection of *BRAF* V600 mutations and *BRAF* fusions by F1CDx and evNGS

	evNGS+	evNGS-	Invalid	Total	
F1CDx+	93	3	0	96	PPV: 96.88% [95%CI*: 91.21%, 98.93%]
F1CDx-	3	102	0	105	NPV: 97.14% [95%CI*: 91.93%, 99.02%]
Invalid	1	1	0	2	
Total	97	106	0	203	
	PPA: 96.88% [95%CI*: 91.21%, 98.93%]	NPA: 97.14% [95%CI*: 91.93%, 99.02%]			

*95% 2-sided confidence intervals (CI) were calculated using the Wilson score method.

There were 6 discordant calls between F1CDx and evNGS test results. Of the 6 calls, the following was observed:

- One sample was detected by F1CDx but was removed due to the contamination QC flag.
- Two samples were detected by the evNGS, but not detected by F1CDx due to limited targeting in the region of the breakpoint.
- Two samples were detected by F1CDx, but not detected by evNGS because *BRAF* intron 7 is not baited on the evNGS assay.
- One sample was detected by F1CDx but not detected by evNGS because the biomarker definition requires the *BRAF* gene to be in the 3' orientation of the fusion product. For that sample, F1CDx detected both orientations, but the evNGS assay only detected the fusion in the biomarker-negative orientation.

i. Analytical Sensitivity

a. Limit of Blank (LoB)

The LoB of zero was confirmed by testing six biomarker negative FFPE glioma tissue samples. Ten replicates for each of the six DNA source samples were processed for a total of 60 sample replicates. Of the 60 replicates, 100% produced valid results and no *BRAF* V600 mutations or *BRAF* fusions were detected across all sample replicates; therefore, the false positive rate for calling alterations was determined to be 0.00%. The study demonstrated that the F1CDx assay does not produce false-positive results in detecting *BRAF* V600 mutations or *BRAF* fusions in biomarker-negative FFPE tissue samples from patients with glioma.

b. Limit of Detection (LoD)

The F1CDx LoD for the detection of *BRAF* V600 mutations and *BRAF* fusions was established through confirmation of platform based LoD and within CDx-specific LoD studies.

The LoD of *BRAF* single nucleotide variant (SNV) alterations was established through a previously conducted study for the PMA approval (P170019) as 3.2% VAF. The LoD was confirmed through a precision study that used samples at or near 1x LoD. The precision study processed at the laboratories in Cambridge, MA (CAM) and Research Triangle Park (RTP) in Morrisville, NC, determined the LoD for *BRAF* SNVs to be 2.49% VAF.

The LoD of *BRAF* fusions was determined through a new LoD study. DNA from two unique glioma samples harboring *BRAF* fusions were processed. Each sample was assessed at five targeted chimeric read levels (30, 24, 18, 12, and 4). Twenty replicates were assessed for each dilution level, except the 30 reads level, which examined 14 replicates. Ninety-four replicates were tested per sample. A total of 188 sample replicates were processed, of which two failed F1CDx QC metrics. The remaining 186 sample replicates passed all QC criteria and were included in the final analysis. Each specimen was evaluated close to the minimum input requirements of the assay (50 ng), thus representing the most challenging evaluation of *BRAF* fusion detection.

The LoD for each sample was determined based on chimeric reads using hit rate method for *BRAF* fusions. The LoD for the two individual *BRAF* fusion samples was 8.30 and 6.75 chimeric reads. The LoD for *BRAF* fusions was determined to be 7.53 chimeric reads, which was the median LoD from the two eligible *BRAF* fusion samples per the hit rate method.

ii. Precision

To evaluate the performance of *BRAF* V600 mutation and *BRAF* fusion detection, a site-to-site precision study was conducted at the laboratories in Cambridge, MA (CAM) and Morrisville, NC (RTP) using 4 unique glioma samples harboring *BRAF* V600 mutations and *BRAF* fusions at challenging DNA input (close to 50 ng) conditions. Table 14 below lists the samples evaluated in this study. Samples were titrated with biomarker negative diluent DNA extracted from glioma FFPE DNA samples to achieve the targeted VAF and chimeric reads (*BRAF* SVs at ~1-1.5x LoD and *BRAF* fusions at ~1-3x LoD). In total, 96 sample replicates (24 replicates of each sample) were evaluated with two reagent lots at each of the two sites.

Table 14. Samples Evaluated in the Precision Study

<i>BRAF</i> -positive samples				<i>BRAF</i> -negative samples	
Sample ID	Disease Ontology	Biomarker-positive Alteration	LoD	Sample ID	Disease Ontology
1	Brain pleomorphic xanthoastrocytoma	<i>CDC42SE2-BRAF</i> fusion	7.53 reads	5	Brain pleomorphic xanthoastrocytoma
2	Brain astrocytoma pilocytic	<i>KIAA1549-BRAF</i> fusion	7.53 reads	6	Brain astrocytoma
3	Brain astrocytoma pilocytic	<i>BRAF</i> 1799T>A/V600E	3.2% VAF	7	Brain astrocytoma pilocytic
4	Brain ganglioglioma	<i>BRAF</i> 1799T>A/V600E	3.2% VAF	8	Brain ganglioglioma

The point estimates and 95% two-sided CIs for reproducibility of each sample at CAM and RTP are detailed in Table 15. Two targeted *BRAF* fusion samples had observed average chimeric reads >1x LoD and were included in the reproducibility results below.

Table 15. Reproducibility for Targeted Variants Above LoD (CAM-RTP)

Source Sample	Targeted Variant	Variant Type	Observed Average Chimeric Reads	LoD (Reads)	# of Positive Replicates	# of Valid Replicates	Reproducibility (%)	95% Two-Sided Score CI (%)	Fold LoD (x)
1	<i>CDC42SE2-BRAF</i>	RE	9.83	7.53	24	24	100.00	[86.20, 100.00]	1.31
2	<i>KIAA1549-BRAF</i>	RE	9.25	7.53	24	24	100.00	[86.20, 100.00]	1.23

Repeatability was evaluated in the two targeted *BRAF* fusions with observed chimeric reads $\geq$ 1x LoD. The point estimates and 95% two-sided CIs for repeatability of each sample are detailed in Table 16.

Table 16. Repeatability for Targeted Variants above LoD (CAM-RTP)

Source Sample	Targeted Variant	# of Agreed Pairs	# of Valid Pairs	Repeatability (%)	95% Two-Sided Score CI (%)
1	<i>CDC42SE2-BRAF</i>	12	12	100.00	[75.75, 100.00]
2	<i>KIAA1549-BRAF</i>	12	12	100.00	[75.75, 100.00]

An exploratory precision analysis was performed for two targeted *BRAF* single variant mutations (Sample ID 3 and 4), since they both had an observed average %VAF <1x LoD. Reproducibility and repeatability results were 100.00% and 100.00%, respectively, for both variants.

B. Animal Studies

No animal studies were conducted using the F1CDx assay.

C. Additional Studies

No additional studies were conducted using the F1CDx assay.

X. SUMMARY OF PRIMARY CLINICAL STUDY

The clinical performance of F1CDx for detecting *BRAF* V600 mutations and *BRAF* fusions in pediatric low-grade glioma (pLGG) patients was demonstrated in a retrospective analysis of specimens from patients enrolled in the FIREFLY-1 clinical study of OJEMDA (tovorafenib). Data generated from the trial supported the clinical validation of the F1CDx assay for identification of patients with *BRAF* V600 mutations and *BRAF* fusions who may benefit from treatment with tovorafenib.

A summary of the clinical study is presented below.

FoundationOne CDx Clinical Bridging Study

A reasonable assurance of safety and effectiveness for F1CDx for detection of *BRAF* V600 mutations and *BRAF* fusions in pLGG patients who may benefit from treatment with OJEMDA (tovorafenib) was demonstrated through a clinical bridging study using tumor tissue FFPE specimens from patients enrolled in the FIREFLY-1 clinical study with known *BRAF* V600 mutation and *BRAF* fusion status, as well as *BRAF* V600 mutation and *BRAF* fusion negative samples that were commercially procured or available from the FMI DNA archives. The clinical bridging study utilized the NovaSeq instrument and CoEx extraction method. The clinical efficacy analysis was performed by analyzing concordance between F1CDx and the enrollment clinical trial assays (CTAs), followed by the imputation of the missing F1CDx results, and determining the clinical outcome of the *BRAF* V600 mutation and *BRAF* fusion positive population identified with F1CDx.

A. **FIREFLY-1 Study Design**

The FIREFLY-1 clinical trial is a Phase 2 multi-center, non-randomized, open-label study of tovorafenib assessing the efficacy, safety, and pharmacokinetics in pediatric patients aged 6 months to ≤ 25 years with relapsed or refractory low-grade glioma (LGGs) or advanced solid tumors harboring a known activating *RAF* alteration, including *BRAF* V600 mutations and *KIAA1549-BRAF* fusions. FIREFLY-1 was initiated on April 22, 2021. This study included three arms:

- Arm 1 (LGG): Patients with recurrent or progressive LGG harboring a known activating *BRAF* alteration, including *BRAF* V600 mutations and *KIAA1549-BRAF* fusions or rearrangements.
- Arm 2 (LGG extension): Patients with recurrent or progressive LGG harboring a known or anticipated activating *RAF* alteration (e.g., *BRAF* or *CRAF/RAF1* fusion or *BRAF* V600 mutation) or rearrangements.
- Arm 3 (advanced solid tumor): Patients with advanced solid tumors harboring a known or anticipated activating *RAF* fusion (e.g., *BRAF* or *CRAF/RAF1* fusion) or rearrangements.

The primary analysis set for the New Drug Application (NDA) population and for the clinical validation studies includes patients enrolled in Arm 1 only. OJEMDA (tovorafenib) was granted accelerated approval by FDA for *BRAF* V600 mutation and *BRAF* fusion positive LGG patients in April 2024 based on data from patients using the Response Assessment in Pediatric Neuro-Oncology (RAPNO)-LGG criteria in the FIREFLY-1 trial. The clinical validation study results presented below follow the same efficacy analysis population used to support the approval of OJEMDA (tovorafenib).

***BRAF* V600 mutation and *BRAF* fusion Evaluation by F1CDx**

The clinical effectiveness of OJEMDA (tovorafenib) among patients with *BRAF* V600 mutations and *BRAF* fusions in pLGG patients as determined by F1CDx was demonstrated in a retrospective analysis of specimens from the FIREFLY-1 clinical trial. A bridging study was conducted to assess: (1) concordance between CTAs and F1CDx in identifying patients with *BRAF* V600 mutations and *BRAF* fusions, (2) the efficacy of OJEMDA (tovorafenib) in patients from the FIREFLY-1 clinical trial who have *BRAF* V600 mutations and *BRAF* fusions as determined by F1CDx and (3) the robustness of the concordance analysis and efficacy analysis with a sensitivity analyses that accounts for the uncertainty due to missing data for *BRAF* V600 mutation and *BRAF* fusion status as determined by F1CDx.

Clinical Bridging Study Design

The clinical bridging study evaluated the clinical validity of F1CDx as a companion diagnostic (CDx) to identify *BRAF* V600 mutation and *BRAF* fusion positive patients from the FIREFLY-1 clinical trial. F1CDx testing was performed on FIREFLY-1 NDA patients who had samples with sufficient tissue material remaining and who

tested positive for *BRAF* V600 mutations and *BRAF* fusions by the CTAs. *BRAF* V600 mutations and *BRAF* fusion negative samples that were commercially procured or from FMI's DNA archives were selected and identified as biomarker negative by an evNGS based assay or a FISH-based assay, and subsequently tested by F1CDx for the CTA negative (CTA-) results.

1. Clinical Inclusion and Exclusion Criteria

Samples meeting pre-defined criteria specified below were included in the clinical bridging study.

Inclusion Criteria for biomarker-positive samples:

- Samples that are derived from LGG FFPE specimens.
- Samples that meet minimum criteria for F1CDx testing requirements.
- *BRAF* alteration-positive samples from the FIREFLY-1 clinical trial that have been obtained with proper informed consent.
- Samples must be obtained in accordance with internal procedures and with Institutional Review Board (IRB) approval.

Inclusion Criteria for negative samples in the NGS cohort:

- Samples must be derived from high confidence LGG FFPE specimens including the following disease ontologies (DOs): brain astrocytoma, brain astrocytoma pilocytic, brain astrocytoma pilomyxoid, brain dysembryonic neuroepithelial tumor, brain gangliocytoma, brain ganglioglioma, and brain oligodendroglioma.
- Samples must be able to support both evNGS and F1CDx DNA input requirements.
- Samples must have $\geq 20\%$ tumor purity.
- Samples must meet the minimum test requirements for the evNGS and F1CDx assays.
- Samples must be obtained in accordance with internal procedures and IRB protocol.

Inclusion Criteria for negative samples in the FISH cohort:

- Samples that are derived from LGG FFPE specimens.
- Samples must meet the minimum test requirements for the FISH and F1CDx assays.
- Samples must be obtained in accordance with internal procedures.

Exclusion Criteria for samples:

- Samples failing to meet any of the inclusion criteria.
- FIREFLY-1 clinical trial samples that lack clear identification or labeling.
- Samples not obtained in accordance with internal procedures or not receiving IRB approval.

- Samples not obtained in accordance with IRB protocol.

2. Follow-up Schedule

The F1CDx clinical bridging study involved only retrospective testing of tissue tumor FFPE samples; as such, no additional patient follow-up was conducted.

3. Clinical Bridging Study Endpoints

The objectives of the F1CDx clinical bridging study were to:

- Establish the clinical validity of F1CDx in identifying *BRAF* V600 mutations and *BRAF* fusions in glioma for treatment with OJEMDA (tovorafenib).
- Evaluate the concordance of results for the *BRAF* V600 mutation and *BRAF* fusion status between the F1CDx assay and the CTAs used for enrollment into the FIREFLY-1 clinical trial.
- Assess the robustness of the concordance and efficacy analyses to account for uncertainty due to missing data for the F1CDx *BRAF* V600 mutation and *BRAF* fusion status.

A concordance analysis was performed for subjects identified with RAPNO-LGG criteria assessed by independent radiology review committee (IRC). Endpoints for the concordance analysis included positive percent agreement (PPA), negative percent agreement (NPA), prevalence adjusted positive predictive value (PPV) and prevalence-adjusted negative predictive value (NPV).

B. Accountability of PMA Cohort

Arm 1 includes the patient population included in the NDA to support accelerated approval of tovorafenib. As of June 2023, 77 patients were enrolled in Arm 1 and were evaluated for the clinical bridging study (*Note: 76 of the 77 patients were available by RAPNO-LGG criteria and is the basis for the approval of OJEMDA (tovorafenib)*). An additional 120 *BRAF* V600 mutation and *BRAF* fusion negative samples were evaluated. Among the 197 samples evaluated, three CTA+ samples did not have available material for testing. A total of 194 samples were processed by F1CDx in this study. Out of 194 samples, 174 samples had valid test results and 20 samples failed during pre- and post- sequencing steps.

Table 17. Sample Accountability for the Clinical Bridging Study by Source Sample

CTA Status	Sample Source	# of Subjects (Samples)	# of F1CDx Unevaluable Samples	# of F1CDx Failed Samples	# of F1CDx Evaluable Samples (%)
Positive	FIREFLY-1 clinical trial	77	3	19	55 (71.43%)
Negative	Samples selected from FMI sample archive and tested negative by evNGS	70	0	0	70 (100.00%)
	Samples commercially procured tested by FISH based assay	24	0	1	23 (95.83%)
	Samples randomly selected from previous validation study that tested negative by evNGS based assay	26	0	0	26 (100.00%)
Total		197	3	20	174 (88.32%)

C. Study Population Demographics and Baseline Parameters

In the clinical bridging study, baseline characteristics were compared between CDx-evaluable and CDx-unevaluable populations of the CTA+ population. The unevaluable population included patients who did not have a sample available to be tested by F1CDx or those whose sample failed F1CDx testing (invalid). The demographics and disease characteristics for the CDx-evaluable and CDx-unevaluable CTA+ patients are provided in Table 18. Differences in characteristics of the CDx-evaluable and CDx-unevaluable populations were identified by covariate analysis. The covariates that showed significant differences between the two groups at a significance level of $\alpha=0.2$ were considered to be imbalanced between the F1CDx-evaluable and the F1CDx-unevaluable groups for patients enrolled based on RAPNO-LGG criteria.

Table 18. Demographic and Clinical Characteristics for CTA+ Patients Identified based on RAPNO-LGG Criteria

Covariate	Group Name	CDx-unevaluable CTA+	CDx-evaluable CTA+	Difference	P-value
Age	Overall	Mean = 8.8, Min = 4, Q1 = 7, Median = 8, Q3 = 10, Max = 16, SD = 3.4	Mean = 9.5, Min = 2, Q1 = 6, Median = 9, Q3 = 12, Max = 21, SD = 4.2	0.7	0.536
Height	Overall	Mean = 133.1, Min = 107.8, Q1 = 122.5, Median = 130.8, Q3 = 139.2, Max = 173, SD = 17	Mean = 135.5, Min = 90.5, Q1 = 121, Median = 137.4, Q3 = 155, Max = 166.6, SD = 20.8	2.4	0.446
Weight	Overall	Mean = 33.3, Min = 14.2, Q1 = 20.7, Median = 29.2, Q3 = 41.8, Max = 78.9, SD = 15.5	Mean = 38.7, Min = 14.6, Q1 = 26.3, Median = 33.8, Q3 = 47.2, Max = 93.4, SD = 17.3	5.5	0.151
BSA	Overall	Mean = 1.1, Min = 0.7, Q1 = 0.8, Median = 1, Q3 = 1.3, Max = 1.9, SD = 0.3	Mean = 1.2, Min = 0.6, Q1 = 1, Median = 1.1, Q3 = 1.4, Max = 2.1, SD = 0.3	0.1	0.194
KPS	Overall	Mean = 95, Min = 90, Q1 = 92.5, Median = 95, Q3 = 97.5, Max = 100, SD = 7.1	Mean = 77.5, Min = 70, Q1 = 70, Median = 75, Q3 = 82.5, Max = 90, SD = 9.6	-17.5	0.153
LPS	Overall	Mean = 95.5, Min = 80, Q1 = 90, Median = 100, Q3 = 100, Max = 100, SD = 6	Mean = 91, Min = 50, Q1 = 90, Median = 90, Q3 = 100, Max = 100, SD = 10.8	-4.5	0.110

Covariate	Group Name	CDx-unevaluable CTA+	CDx-evaluable CTA+	Difference	P-value
Time since specimen acquisition to clinical trial enrollment	Overall	Mean = 50.5, Min = 0.3, Q1 = 29, Median = 56.7, Q3 = 78.1, Max = 92.2, SD = 31.3	Mean 44.6, Min = 0.1, Q1 = 8.3, Median = 34.8, Q3 = 68.8, Max = 157.6, SD = 41.1	-6	0.370
Age Group	Overall	22	53	31	0.311
	12 years to <16 years	2(9.1%)	14(26.4%)	12(17.3%)	
	16 years to <=25 years	2(9.1%)	4(7.5%)	2(-1.5%)	
	2 years to <6 years	3(13.6%)	10(18.9%)	7(5.2%)	
	6 years to <12 years	15(68.2%)	25(47.2%)	10(-21.0%)	
Race	Overall	22	53	31	0.801
	ASIAN	2(9.1%)	3(5.7%)	1(-3.4%)	
	BLACK OR AFRICAN AMERICAN	1(4.5%)	1(1.9%)	0(-2.7%)	
	MULTIPLE	0(0.0%)	3(5.7%)	3(5.7%)	
	OTHER	1(4.5%)	4(7.5%)	3(3.0%)	
	WHITE	13(59.1%)	27(50.9%)	14(-8.1%)	
	N/A	5(22.7%)	15(28.3%)	10(5.6%)	
Race Group	Overall	22	53	31	0.754
	Others	4(18.2%)	11(20.8%)	7(2.6%)	
	White	13(59.1%)	27(50.9%)	14(-8.1%)	
	N/A	5(22.7%)	15(28.3%)	10(5.6%)	
Ethnicity	Overall	22	53	31	1.000
	HISPANIC OR LATINO	0(0.0%)	2(3.8%)	2(3.8%)	

Covariate	Group Name	CDx-unevaluable CTA+	CDx-evaluable CTA+	Difference	P-value
	NOT HISPANIC OR LATINO	17(77.3%)	33(62.3%)	16(-15.0%)	
	N/A	5(22.7%)	18(34.0%)	13(11.2%)	
Sex	Overall	22	53	31	0.458
	F	9(40.9%)	27(50.9%)	18(10.0%)	
	M	13(59.1%)	26(49.1%)	13(-10.0%)	
Primary Tumor Site	Overall	22	53	31	0.480
	Brain Stem	3(13.6%)	3(5.7%)	0(-8.0%)	
	Cerebellum	0(0.0%)	5(9.4%)	5(9.4%)	
	Cerebral Hemisphere	2(9.1%)	3(5.7%)	1(-3.4%)	
	Deep Midline Structures	3(13.6%)	6(11.3%)	3(-2.3%)	
	Optical Pathway	12(54.5%)	26(49.1%)	14(-5.5%)	
	Other	2(9.1%)	10(18.9%)	8(9.8%)	
Disease ontology subtype	Overall	22	53	31	0.701
	Astrocytic	20(90.9%)	50(94.3%)	30(3.4%)	
	Mixed Glial-neuronal	2(9.1%)	2(3.8%)	0(-5.3%)	
	Other	0(0.0%)	1(1.9%)	1(1.9%)	
Pre-operative staging	Overall	22	53	31	1.000
	Disseminated/Metastatic Disease	2(9.1%)	6(11.3%)	4(2.2%)	
	Leptomeningeal Spread	2(9.1%)	6(11.3%)	4(2.2%)	
	Localized Disease	18(81.8%)	41(77.4%)	23(-4.5%)	
	Overall	22	53	31	0.726

Covariate	Group Name	CDx-unevaluable CTA+	CDx-evaluable CTA+	Difference	P-value
Post-operative staging	Biopsy Only, Not Attempted	13(59.1%)	26(49.1%)	13(-10.0%)	
	Gross Total Resection	0(0.0%)	1(1.9%)	1(1.9%)	
	Sub-Total Resection	9(40.9%)	26(49.1%)	17(8.1%)	
Prior <i>BRAF</i> inhibitor therapy	Overall	22	53	31	1.000
	N	20(90.9%)	48(90.6%)	28(-0.3%)	
	Y	2(9.1%)	5(9.4%)	3(0.3%)	
Prior MEK inhibitor therapy	Overall	22	53	31	0.321
	N	12(54.5%)	22(41.5%)	10(-13.0%)	
	Y	10(45.5%)	31(58.5%)	21(13.0%)	
Number of prior lines of systemic therapies [†]	Overall	22	53	31	0.971
	1	5(22.7%)	12(22.6%)	7(-0.1%)	
	2	6(27.3%)	13(24.5%)	7(-2.7%)	
	3	5(22.7%)	11(20.8%)	6(-2.0%)	
	4	3(13.6%)	6(11.3%)	3(-2.3%)	
	more than 5	3(13.6%)	11(20.8%)	8(7.1%)	

[†] More than 5 prior lines of therapies were aggregated into one group to increase number of data points within groups.

D. Safety and Effectiveness Results

1. Safety Results

The safety with respect to treatment with tovorafenib was addressed during the review of the tovorafenib NDA (No. 218033 and No. 21770) and is not addressed in detail in this Summary of Safety and Effectiveness Data. The evaluation of safety was based on the analysis of adverse events (AEs), clinical laboratory evaluations, physical examinations, and vital signs. Please refer to Drugs@FDA for complete safety information on OJEMDA (tovorafenib).

No adverse events were reported in the conduct of the clinical bridging study used to support this PMA supplement, as the study was performed retrospectively using banked samples.

2. Effectiveness Results

The effectiveness of F1CDx to identify patients with *BRAF* V600 mutations and *BRAF* fusions who may benefit from treatment with tovorafenib is supported by the evaluation of LGG patient population enrolled into the FIREFLY-1. The concordance results, efficacy analysis and sensitivity analysis of missing results are presented for the RAPNO-LGG evaluable population.

a. Concordance Results

A concordance analysis between CTAs and F1CDx was performed with the 75 (excluding the one F1CDx-enrolled patient) RAPNO-LGG evaluable patients with valid F1CDx testing results and 120 CTA- samples. A contingency table is provided below (Table 19).

Table 19. Contingency Table Comparing *BRAF* V600 Mutation or *BRAF* Fusion Between the CTA and F1CDx Based on RAPNO-LGG Evaluable Population

	CTA		
	CTA+	CTA-	Total
F1CDx+	41	2	43
F1CDx-	12	117	129
F1CDx-unevaluable	22	1	23
Total	75	120	195

PPA was established as 77.36% (95% two-sided score CI: [64.47%, 86.55%]) and NPA was 98.32% (95% two-sided score CI: [94.08%, 99.54%]), as shown in Table 20. Prevalence adjusted PPV and NPV were calculated using 70.96% as the biomarker prevalence^{9,10,11}. The prevalence adjusted PPV was 99.12% (95% two-sided CI: [97.77%, 100.00%]) and the prevalence adjusted NPV was 63.99% (95% two-sided CI: [53.79%, 77.60%]), as shown in Table 20.

Table 20. Concordance Results for *BRAF* V600 Mutation or *BRAF* Fusion with F1CDx Relative to CTA Based on the RAPNO-LGG Evaluable Population

Metric	Numerator	Denominator	Point estimate (%)	95% Two-sided CI (%)	Method for CI
PPA	41	53	77.36	[64.47, 86.55]	Wilson score
NPA	117	119	98.32	[94.08, 99.54]	Wilson score
Adjusted PPV	N/A	N/A	99.12	[97.77, 100.00]	Bootstrap
Adjusted NPV	N/A	N/A	63.99	[53.79, 77.60]	Bootstrap

The 12 discordant results seen in the CTA+ population had evidence of low tumor content which can impact the sensitivity of F1CDx for detection of somatic alterations, including fusions.

b. Clinical Efficacy Results

The clinical validity of F1CDx for the detection of *BRAF* V600 mutations and *BRAF* fusions in pLGG patients for the CTA+ population was based on estimation of clinical efficacy in the F1CDx-positive population and subgroups of the CTA+ population stratified by F1CDx status. The efficacy endpoint was overall response rate (ORR) as assessed by IRC based on the RAPNO-LGG evaluable population. ORR was defined as the observed proportion of patients whose best overall response is confirmed complete response (CR) or partial response (PR) as determined by IRC and by the treating investigator. The point estimate ORR and two-sided 95% CI, calculated using the Wilson-Score method, for each sub-population is presented in Table 21 below. An ORR of 52.00% (39/75, 95% two-sided score CI: [40.87%, 62.93%]) was observed for the 75 CTA-enrolled biomarker-positive subjects based on RAPNO-LGG criteria. Since the adjusted PPV was not 100%, the tipping point analysis under varied values of the tuning parameter (denoted as *c*) was performed to estimate the ORR for patients identified as biomarker positive by F1CDx (F1CDx+). The estimated F1CDx ORR ranged from 50.95% to 51.22% when *c* ranged from 0 to 1. The ORR and two-sided 95% CI for F1CDx+ population is presented in Table 22. These results showed that the ORR for the F1CDx+ population is comparable to the ORR for the CTA+ population, to demonstrate the clinical validity of F1CDx in identifying *BRAF* alterations V600 mutations and *BRAF* fusions in the RAPNO-LGG evaluable population who may benefit from the treatment of tovorafenib.

Table 21. Efficacy (ORR as determined by IRC using the RAPNO-LGG Criteria) based on all NDA Patients

Sub-population	No. of Patients	No. of Responders	ORR (%)	95% Two-sided Wilson Score CI (%) *
CTA+	75	39	52.00	[40.87, 62.93]
F1CDx+ CTA+	41	21	51.22	[36.48, 65.75]
F1CDx- CTA+	12	5	41.67	[19.33, 68.05]
F1CDx-evaluable CTA+	53	26	49.06	[36.12, 62.12]
F1CDx-unevaluable CTA+	22	13	59.09	[38.73, 76.74]
F1CDx-enrolled	1	0	0.00	N/A

*CIs are only reported if the sample size > 10.

Table 22. Bridging study estimated drug efficacy (ORR as determined by IRC using the RAPNO-LGG evaluable population) for the F1CDx+ population based on observed data

<i>c</i>	Estimated ORR for F1CDx+ (%)	95% two-sided Wald CI (%)*
0	50.95	[35.78, 66.13]
0.3	51.03	[35.86, 66.20]
0.5	51.09	[35.92, 66.26]
0.7	51.14	[35.97, 66.31]
1.0	51.22	[36.05, 66.39]

*CIs are only reported if the sample size >10.

c. Sensitivity Analysis for Missing CDx Results

A sensitivity analysis was conducted to assess the robustness of the concordance and the clinical efficacy results accounting for the patients with missing F1CDx status. Samples were considered unevaluable if the samples were not tested or if they were tested but returned as invalid result.

The sensitivity analysis employed the multiple imputation method to impute missing F1CDx results and used the imputed and observed datasets to estimate the concordance and the ORR for F1CDx positive patients. Because the prevalence adjusted PPV is not 100% in the sensitivity analysis, the tipping point approach for the point estimate of the ORR for the F1CDx+ population was also conducted for the sensitivity analysis. The mean ORR estimated for the F1CDx+ population ranges from 51.46% to 51.73% when *c* ranges from 0 to 1, which is comparable to the efficacy estimates for F1CDx+ patients (ranges from 50.95% to 51.22% when *c* ranges from 0 to 1) in the observed data. Overall, the sensitivity analysis results demonstrated the robustness of the concordance and the clinical efficacy results accounting for the missing F1CDx statuses.

3. Subgroup Analyses

The ORR efficacy results by subgroup for *BRAF* V600 mutations and *BRAF* fusions, are presented in Tables 23 and 24 below. In general, the ORR for the F1CDx+|CTA+ population is comparable to the ORR for the CTA+ population, which demonstrates the clinical utility of F1CDx in identifying *BRAF* V600 mutation and *BRAF* fusion-positive LGG patients by RAPNO-LGG criteria who may respond to tovorafenib.

Table 23. Efficacy (ORR as determined by IRC using the RAPNO-LGG Criteria) based on Patients Harboring *BRAF* V600 Mutations

Sub-Population	No. of Patients	No. of CR/PR	ORR (%)	95% Two-Sided Wilson Score CI (%)*
CTA+	12	6	50.00	[25.38, 74.62]
F1CDx+ CTA+	9	5	55.56	N/A
F1CDx- CTA+	0	N/A	N/A	N/A
F1CDx-evaluable CTA+	9	5	55.56	N/A
F1CDx-unevaluable [†] CTA+	3	1	33.33	N/A
F1CDx enrolled	0	N/A	N/A	N/A

* CIs are only reported if the sample size is >10.

[†] Patients were considered F1CDx unevaluable if the sample failed QC or didn't have sufficient material for F1CDx testing.

Table 24. Efficacy (ORR as determined by IRC using the RAPNO-LGG Criteria) based on Patients Harboring *BRAF* Fusions

Sub-Population	No. of Patients	No. of CR/PR	ORR (%)	95% Two-sided Wilson Score CI (%)*
CTA+	55	29	52.73	[39.79, 65.31]
F1CDx+ CTA+	28	14	50.00	[32.63, 67.37]
F1CDx- CTA+	9	3	33.33	N/A
F1CDx-evaluable CTA+	37	17	45.95	[31.04, 61.62]
F1CDx-unevaluable ^a CTA+	18	12	66.67	[43.75, 83.72]
F1CDx-enrolled	1	0	0.00	N/A

*CIs are only reported if the sample size > 10.

^a Patients were considered F1CDx-unevaluable if the sample failed QC or didn't have sufficient material for F1CDx testing.

Overall, there were 9 out of 37 (24.3%) F1CDx-evaluable|CTA+ patients that received a negative F1CDx result for *BRAF* fusions, i.e., F1CDx-|CTA+. Of the 9 discordant patients, 3 patients had a complete or partial response to OJEMDA (tovorafenib) treatment, supporting that these 3 samples were likely true positive samples. Investigation into the discordances showed that the reduced sensitivity to detect *BRAF* fusions in these samples was attributed to sub-optimal quality metrics, including low tumor purity, and limited targeting of *BRAF* intron 8 in the F1CDx baits. Since F1CDx failed to detect 24.3% of *BRAF* fusions and because these patients may forgo a potentially effective therapy, a limitation stating that F1CDx may miss a subset of patients with *BRAF* fusions who may derive benefit from tovorafenib is included in the device labeling.

4. Pediatric Extrapolation

In this premarket application, the safety and effectiveness of the proposed device, F1CDx, is supported by the clinical data from clinical trial FIREFLY-1 in the pediatric patients 6 months of age and older with relapsed or refractory pediatric low-grade glioma (LGG) harboring a *BRAF* V600 mutations and *BRAF* fusions who may benefit from treatment with OJEMDA (tovorafenib). The safety and effectiveness of OJEMDA (tovorafenib) in pediatric patients 6 months of age and older with relapsed or refractory pediatric LGG harboring a *BRAF* fusion or *BRAF* V600 mutation have been established based on data from a multicenter, open-label, single-arm clinical trial, as demonstrated in the drug label. In this clinical trial, the median age of patients was 9 years (range 1 to 24 years).

Low-grade gliomas are the most common brain tumor diagnosed in children (Ostrom 2019). They can occur anywhere in the central nervous system (CNS) and can comprise multiple different tumor histology. Pediatric LGGs comprise 30% of all childhood brain tumors with an annual incidence of 1,000 to 1,600 in the US (Diwanji 2017). Using SEER data from 2017, the complete prevalence of *BRAF*-altered LGG subtypes in the US in 2019 was estimated at 7,813 cases among individuals aged 0-18 years.

One of the most frequently altered genes in the MAPK pathway is v-raf murine sarcoma viral oncogene homolog B (*BRAF*). *BRAF* alterations are the most common oncogenic drivers in pediatric low-grade glioma and occur in up to 70% of pediatric low-grade glioma (Ryall 2020a; Ryall 2020b; Faulkner 2015). These genomic alterations result in increased *BRAF* kinase activity that is independent of extracellular stimuli or rat sarcoma virus (RAS)-GTP as well as constitutive signaling that drives cell proliferation and tumor growth (Davies 2002; Jones 2008).

The most common *BRAF* alteration found in pediatric low-grade glioma is the KIAA1549:*BRAF* fusion. The KIAA1549:*BRAF* fusion is found in approximately 65% of *BRAF*-altered pediatric low-grade glioma. Although rare, other oncogenic *BRAF* fusions have been identified in pediatric low-grade glioma, all of which consist of an N-terminal dimerization domain from another protein and the C-terminal kinase domain of *BRAF*. The canonical *BRAF* V600E mutation is found in approximately 30% of *BRAF*-altered pediatric low-grade glioma. (Ryall 2020b; Schreck 2019; Schreck 2023; Zwaig 2022).

XI. FINANCIAL DISCLOSURE

The Financial Disclosure by Clinical Investigators regulation (21 CFR 54) requires applicants who submit a marketing application to include certain information concerning the compensation to, and financial interests and arrangement of, any clinical investigator conducting clinical studies covered by the regulation. The pivotal clinical study included two investigators of which two were full-time or part-time employees of the sponsor and

two of the investigators had disclosable financial interests/arrangements as defined in 21 CFR 54.2(a), (b), (c) and (f) and described below:

- Compensation to the investigator for conducting the study where the value could be influenced by the outcome of the study: [0]
- Significant payment of other sorts: [0]
- Proprietary interest in the product tested held by the investigator: [1]
- Significant equity interest held by investigator in sponsor of covered study: [0]

The applicant has adequately disclosed the financial interest/arrangements with clinical investigators. Statistical analyses were conducted by FDA to determine whether the financial interests/arrangements had any impact on the clinical study outcome. The information provided does not raise any questions about the reliability of the data.

XII. PANEL MEETING RECOMMENDATION AND FDA'S POST-PANEL ACTION

In accordance with the provisions of section 515(c)(3) of the act as amended by the Safe Medical Devices Act of 1990, this PMA supplement was not referred to the Molecular and Clinical Genetics Panel, an FDA advisory committee, for review and recommendation because the information in the PMA substantially duplicates information previously reviewed by this panel.

XIII. CONCLUSIONS DRAWN FROM PRECLINICAL AND CLINICAL STUDIES

A. Effectiveness Conclusions

The effectiveness of F1CDx to identify glioma patients with *BRAF* V600 mutations and *BRAF* fusions who may benefit from OJEMDA (tovorafenib) treatment is supported by the results from the clinical validation study. This study was performed using specimens from patients enrolled in the FIREFLY-1 clinical trial with known *BRAF* V600 mutation and *BRAF* fusion status and supplemented with *BRAF* V600 mutation and *BRAF* fusion negative samples that were commercially procured or selected from the FMI archives. The data from the analytical validation and clinical bridging studies support the reasonable assurance of safety and effectiveness of the F1CDx assay when used in accordance with the indications for use. Data from the FIREFLY-1 trial show that patients who had *BRAF* V600 mutations and *BRAF* fusions received benefit from treatment with OJEMDA (tovorafenib) and support the addition of the CDx indication to F1CDx. The bridging study demonstrated the ability of F1CDx to detect *BRAF* V600 mutation and *BRAF* fusion positive patients that benefit from OJEMDA (tovorafenib) therapy.

B. Safety Conclusions

The risks of the device are based on data collected in the analytical studies conducted to support PMA approval as described above. The F1CDx assay is an in vitro diagnostic test, which involves testing of DNA extracted from FFPE tumor tissue. The assay can be performed using DNA extracted from existing (archival) tissue samples routinely collected as part of the diagnosis and patient care.

Failure of the device to perform as expected or failure to correctly interpret test results may lead to incorrect test results, and subsequently, inappropriate patient management decisions in cancer treatment. Patients with false positive results may undergo treatment with one of the therapies listed in Table 1 of the intended use statement without clinical benefit and may experience adverse reactions associated with the therapy. Patients with false negative results may not be considered for treatment with the indicated therapy. There is also a risk of delayed results, which may lead to delay of treatment with the indicated therapy.

C. Benefit-Risk Determination

The probable benefits of F1CDx in identification of *BRAF* V600 mutations and *BRAF* fusions for treatment with OJEMDA (tovorafenib) are based on data collected in the FIREFLY-1 clinical trial and the clinical bridging study. The clinical benefit of the F1CDx assay for the selection of glioma patients with *BRAF* V600 mutations and *BRAF* fusions was demonstrated in a retrospective bridging study using samples from patients enrolled in the FIREFLY-1 clinical study and supplemented with additional *BRAF* V600 mutation and *BRAF* fusion negative samples. Clinical outcome of *BRAF* V600 mutation and *BRAF* fusion positive patients by F1CDx, as assessed by IRC using RAPNO-LGG criteria, indicated an ORR range from 50.95% to 51.22%, which was comparable to the ORR in the CTA+ population and provides a meaningful clinical benefit in this population. This supports the probable benefit of F1CDx in selecting *BRAF* V600 mutation and *BRAF* fusion positive patients for treatment with OJEMDA (tovorafenib).

There is potential risk associated with the use of this device, mainly due to 1) false positive, false negatives, or failure to provide a result, and 2) incorrect interpretation of test results by the user. The risks of the F1CDx assay are associated with the potential mismanagement of patients resulting from false results of the test. Patients who are determined to be false positive by the test may be exposed to OJEMDA (tovorafenib), a drug that may lead to adverse events without probable benefit or may have delayed access to treatments that could be more beneficial. A false negative result may prevent a patient from accessing OJEMDA (tovorafenib), which has demonstrated clinically meaningful benefit to patients who carry *BRAF* V600 mutations or *BRAF* fusions.

The risk of false results is partially mitigated by clinical and analytical studies presented above. Based on data from the clinical trial, there is a risk that a subset of patients with *BRAF* V600 mutations and *BRAF* fusions who may benefit from OJEMDA

(tovorafenib) may be missed by F1CDx. In addition, analytical concordance study of F1CDx to clinical trial assays (CTAs) used for enrollment, revealed, a PPA of 77.36% (41/53, 95% CI: 64.47%, 86.55%) and NPA was 98.2% (117/119, 95% CI: 94.08%, 99.54%), for the detection of *BRAF* V600 mutations and *BRAF* fusions, conditional on the CTA. This data reveals a risk of false negativity with this device. To further address this, we incorporated the following limitation that further serve to mitigate the risks of this test.

This assay did not detect 24.3% (9/37) of *BRAF* fusions in the clinical validation study supporting this indication. Reduced sensitivity to detect *BRAF* fusions in the clinical validation samples was attributed to sub-optimal quality metrics, including low tumor purity, observed across the whole sample population. Physicians who receive a negative report for *BRAF* fusions for patients with pediatric low-grade glioma should consider following up with another test such that patients may not forgo a potentially effective therapy.

The clinical and analytical performance of the device included in this submission demonstrate that the assay is expected to perform with reasonable accuracy, partially mitigating the potential for false results. In addition, to supplement the premarket data, some post-market studies are planned as summarized in Section XIV, below.

1. Patient Perspective

This submission either did not include specific information on patient perspectives or the information did not serve as part of the basis of the decision to approve or deny the PMA for this device.

In conclusion, given the available information above, the data support that for the F1CDx assay, and the indications noted in the intended use statement, the probable benefits outweigh the probable risks.

D. Overall Conclusions

The data in this application support the reasonable assurance of safety and effectiveness of this device when used in accordance with the indications for use. Data from the analytical and clinical studies support the performance of F1CDx as an aid for the identification of *BRAF* V600 mutations and *BRAF* fusions in pediatric LGG patients for whom tovorafenib may be indicated.

XIV. CDRH DECISION

CDRH issued an approval order for the PMA (P170019/S054) on 1/16/2025.

Foundation Medicine, Inc. (FMI) must provide the following software information in a post-approval report within 6 months of approval of this PMA supplement:

1. FMI must submit a list of the cumulative changes and in sufficient detail acceptable to FDA, made between the currently deployed genomics platform, which includes analysis pipeline (AP) software version v3.37.0 or the most current approved version at the time of submitting the study report to FDA, and the AP versions used in the analytical and clinical validation studies in this PMA supplement.
2. FMI must submit a detailed description of the validation activity conducted to support the version change, including the associated risk assessments for each change, and the rationale, acceptable to FDA, that the validation performed supports reasonable assurance that the modification has not affected the performance or raised new concerns regarding the safety and effectiveness of the device.
3. FMI must provide evidence, acceptable to FDA, that performance expectations with the currently deployed genomics platform, including AP v3.37.0 or the most current approved version at the time of submitting the study report to FDA, are representative of the performance observed in the analytical and clinical validation studies conducted with the previous AP versions (v3.22.0 to v3.33.0) in this PMA supplement. Such evidence may include regression testing using the clinical and analytical datasets to perform in silico reanalysis of the results obtained in the validation studies and confirmation that there is little or no deviation in the quality metrics for each of the samples to support that the performance of the assay remains the same.

XV. APPROVAL SPECIFICATIONS

Directions for use: See device labeling.

Hazards to Health from Use of the Device: See Indications, Contraindications, Warnings, Precautions, and Adverse Events in the device labeling.

Post-approval Requirements and Restrictions: See approval order.

XVI. REFERENCES

1. Fisher S., Barry A., Abreu J., et al. A scalable, fully automated process for construction of sequence-ready human exome targeted capture libraries. *Genome Biol* 12, R1 (2011).
2. Karolchik, D., Hinrichs AS, Kent WJ., et al. The UCSC Table Browser data retrieval tool. *Nucleic Acids Res* 32, D493-496 (2004).
3. Gnirke A., Melnikov A., Maguire J., et al. Solution hybrid selection with ultra-long oligonucleotides for massively parallel targeted sequencing. *Nat Biotechnol* 27, 182-189 (2009).
4. Li H. & Durbin, R. Fast and accurate long-read alignment with Burrows-Wheeler transform. *Bioinformatics* 26, 589-595 (2010).
5. Li H., Handsaker B., Wysoker A., et al. The Sequence Alignment/Map format and SAMtools. *Bioinformatics* 25, 2078-2079 (2009).

6. DePristo MA., Banks E., Poplin R., et al. A framework for variation discovery and genotyping using next-generation DNA sequencing data. *Nat Genet* 43, 491-498 (2011).
7. Forbes SA, Bindal N., Bamford S., et al. COSMIC: mining complete cancer genomes in the Catalogue of Somatic Mutations in Cancer. *Nucleic Acids Res* 39, D945-950 (2011).
8. Compeau PE., Pevzner PA., Tesler, G. How to apply de Bruijn graphs to genome assembly. *Nat Biotechnol* 29, 987-991 (2011).
9. FIREFLY-1: A Phase 2, Open-Label, Multicenter Study to Evaluate the Safety and Efficacy of the Oral Pan-RAF Inhibitor DAY101 in Pediatric Patients with RAF-Altered, Recurrent or Progressive Low-Grade Glioma and Advanced Solid Tumors, 21 October 2021
10. Penman CL, Faulkner C, Lowis SP, et al. Current understanding of *BRAF* alterations in diagnosis, prognosis, and therapeutic targeting in pediatric low-grade gliomas. *Front Oncol.* 2015;5:54.
11. Behling F, Schittenhelm J. Oncogenic *BRAF* alterations and their role in brain tumors. *Cancers.* 2019;11(6):794.
12. Davies H, Bignell GR, Cox C, et al. Mutations of the *BRAF* gene in human cancer. *Nature.* 2002;417(6892):949-54.
13. Diwanji TP, Engelman A, Snider JW, et al. Epidemiology, diagnosis, and optimal management of glioma in adolescents and young adults. *Adolesc Health Med Ther.* 2017;8:99-113.
14. Faulkner C, Ellis HP, Shaw A, et al. *BRAF* fusion analysis in pilocytic astrocytomas: KIAA1549*BRAF* 15-9 fusions are more frequent in the midline than within the cerebellum. *J Neuropathol Exp Neurol.* 2015 Sep;74(9):867-72.
15. Jones DT, Kocialkowski S, Liu L, et al. Tandem duplication producing a novel oncogenic *BRAF* fusion gene defines the majority of pilocytic astrocytomas. *Cancer Res.* 2008;68(21):8673-8677.
16. Ostrom QT, Adel Fahmideh M, Cote DJ, et al. Risk factors for childhood and adult primary brain tumors. *Neuro Oncol.* 2019;21(11):1357-1375.
17. Ryall S, Tabori U, Hawkins C. Pediatric low-grade glioma in the era of molecular diagnostics. *Acta Neuropathol Commun.* 2020a;8(1):30.
18. Ryall S, Zapotocky M, Fukuoka K, Nobre L, Stucklin AG, Bennett J, et al. Integrated molecular and clinical analysis of 1,000 pediatric low-grade gliomas. *Cancer Cell* 2020b; 37:569-83.
19. Schreck KC, Grossman SA, Pratilas CA. *BRAF* mutations and the utility of RAF and MEK inhibitors in primary brain tumors. *Cancers (Basel).* 2019; 11(9):1262.